

FortiOS™ Handbook - Hardware Acceleration

VERSION 5.0

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FortiOS™ Handbook - Hardware Acceleration

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Change Log

Date	Change Description
14 September 2015	Corrected the <code>diagnose npu np6 port-list</code> output for the FortiGate-1500D (see FortiGate-1500D fast path architecture on page 43).
8 September 2015	Corrected the CLI commands in FortiGate-3700D low latency fast path architecture on page 50 and FortiGate-3700D normal latency fast path architecture on page 52 .
30 August 2015	New chapter Hardware acceleration diagnose commands on page 88 .
	Consolidated all information about link aggregation groups (LAGs) and NP6 and NP4 offloading into the following sections: Increasing NP6 offloading capacity using link aggregation groups (LAGs) on page 24 and Increasing NP4 offloading capacity using link aggregation groups (LAGs) on page 60 .
	Also removed incorrect information about LAGs from other sections in this document and from the FortiOS IPsec VPN handbook.
26 August 2015	Corrected errors with the descriptions of FortiGate-1240B port mapping in FortiGate-1240B on page 76 .
	Corrected errors about configuring the FortiGate-3700D for low latency and how NP6 port mapping is affected in FortiGate-3700D fast path architecture on page 50 .
	New hardware platforms added: FortiGate-300D fast path architecture on page 33 , FortiGate-400D fast path architecture on page 34 , FortiGate-600D fast path architecture on page 36 , and FortiGate-900D fast path architecture on page 37 .
4 March 2015	Updated 5.0 version.

Introduction

This document describes the hardware components that Fortinet builds into FortiGate devices to accelerate traffic through FortiGate units. Three types of hardware acceleration components are described:

- Content processors (CPs) that accelerate a wide range of security functions
- Security processors (SPs) that accelerate specific security functions
- Network processors (NPs) that offload network traffic to specialized hardware that is optimized to provide high levels of network throughput.

This FortiOS Handbook chapter contains the following sections:

[Hardware acceleration overview](#) describes the capabilities of FortiGate content processors (CPs), security processors (SPs) and network processors (NPs). This chapter also describes how to determine the hardware acceleration components installed in your FortiGate unit and contains some configuration details and examples.

[NP6 Acceleration](#) describes the FortiGate NP6 network processor.

[FortiGate NP6 architectures](#) contains details about the network processing architectures of FortiGate units that contain NP6 processors.

[NP4 Acceleration](#) describes the FortiGate NP4 network processor.

[FortiGate NP4 architectures](#) contains details about the network processing architectures of FortiGate units that contain NP4 processors.

Hardware acceleration overview

Most FortiGate models have specialized acceleration hardware that can offload resource intensive processing from main processing (CPU) resources. Most FortiGate units include specialized content processors (CPs) that accelerate a wide range of important security processes such as virus scanning, attack detection, encryption and decryption. (Only selected entry-level FortiGate models do not include a CP processor.) Many FortiGate models also contain security processors (SPs) that accelerate processing for specific security features such as IPS and network processors (NPs) that offload processing of high volume network traffic.

Content processors (CP4, CP5, CP6 and CP8)

Most FortiGate models contain FortiASIC Content Processors (CPs) that accelerate many common resource intensive security related processes. CPs work at the system level with tasks being offloaded to them as determined by the main CPU. Capabilities of the CPs vary by model. Newer FortiGate units include CP8 processors. Older CP versions still in use in currently operating FortiGate models include the CP4, CP5, and CP6.

CP8 capabilities

The CP8 content processor provides the following services:

- IPS signature matching acceleration
- High performance VPN bulk data engine
 - IPsec and SSL/TLS protocol processor
 - DES/3DES/AES in accordance with FIPS46-3/FIPS81/FIPS197
 - ARC4 in compliance with RC4
 - MD5/SHA-1/SHA256 with RFC1321 and FIPS180
 - HMAC in accordance with RFC2104/2403/2404 and FIPS198
- Key Exchange Processor support high performance IKE and RSA computation
 - Public key exponentiation engine with hardware CRT support
 - Primarily checking for RSA key generation
 - Handshake accelerator with automatic key material generation
 - Random Number generator compliance with ANSI X9.31
 - Sub public key engine (PKCE) to support up to 4096 bit operation directly
- Message authentication module offers high performance cryptographic engine for calculating SHA256/SHA1/MD5 of data up to 4G bytes (used by many applications)
- PCI express Gen 2 four lanes interface
- Cascade Interface for chip expansion

CP6 capabilities

- Dual content processors
- FIPS-compliant DES/3DES/AES encryption and decryption
- SHA-1 and MD5 HMAC with RFC1321 and FIPS180

- HMAC in accordance with RFC2104/2403/2404 and FIPS198
- IPsec protocol processor
- High performance IPsec engine
- Random Number generator compliance with ANSI X9.31
- Key exchange processor for high performance IKE and RSA computation
- Script Processor
- SSL/TLS protocol processor for SSL content scanning and SSL acceleration

CP5 capabilities

- FIPS-compliant DES/3DES/AES encryption and decryption
- SHA-1 and MD5 HMAC with RFC1321/2104/2403/2404 and FIPS180/FIPS198
- IPsec protocol processor
- High performance IPSEC Engine
- Random Number generator compliant with ANSI X9.31
- Public Key Crypto Engine supports high performance IKE and RSA computation
- Script Processor

CP4 capabilities

- FIPS-compliant DES/3DES/AES encryption and decryption
- SHA-1 and MD5 HMAC
- IPSEC protocol processor
- Random Number generator
- Public Key Crypto Engine
- Content processing engine
- ANSI X9.31 and PKCS#1 certificate support

Determining the content processor in your FortiGate unit

Use the `get hardware status` CLI command to determine which content processor your FortiGate unit contains. The output looks like this:

```
get hardware status
Model name: FortiGate-100D
ASIC version: CP8
ASIC SRAM: 64M
CPU: Intel(R) Atom(TM) CPU D525 @ 1.80GHz
Number of CPUs: 4
RAM: 1977 MB
Compact Flash: 15331 MB /dev/sda
Hard disk: 15272 MB /dev/sda
USB Flash: not available
Network Card chipset: Intel(R) PRO/1000 Network Connection (rev.0000)
Network Card chipset: bcm-sw Ethernet driver 1.0 (rev.)
```

The ASIC version line lists the content processor model number.

Viewing SSL acceleration status

You can view the status of SSL acceleration using the following command:

```
get vpn status ssl hw-acceleration-status
Acceleration hardware detected: kxp=on cipher=on
```

Disabling CP offloading for firewall policies

If you want to completely disable offloading to CP processors for test purposes or other reasons, you can do so in security policies. Here are some examples:

For IPv4 security policies.

```
config firewall policy
  edit 1
    set auto-asic-offload disable
  end
```

For IPv6 security policies.

```
config firewall policy6
  edit 1
    set auto-asic-offload disable
  end
```

For multicast security policies.

```
config firewall multicast-policy
  edit 1
    set auto-asic-offload disable
  end
```



Disabling `auto-asic-offload` also disables NP offloading.

Security processors (SPs)

FortiGate Security Processing (SP) modules, such as the SP3 but also including the XLP, XG2, XE2, FE8, and CE4, work at both the interface and system level to increase overall system performance by accelerating specialized security processing. You can configure the SP to favor IPS over firewall processing in hostile high-traffic environments.

SP processors include their own IPS engine which is similar to the FortiOS IPS engine but with the following limitations:

- The SP IPS engine does not support SSL deep inspection. When you have SSL deep inspection enabled for a security policy that includes flow-based inspection or IPS, offloading to the SP is disabled and traffic is processed by the FortiGate CPU and CP processors.

- The SP IPS engine does not support FortiGuard Web Filtering. When you enable flow-based FortiGuard Web Filtering on a FortiGate unit with an SP processor, the SP processor cannot perform FortiGuard lookups and web pages fail to load.

The following security processors are available:

- The SP3 (XLP) is built into the FortiGate-5101B and provides IPS acceleration. No special configuration is required. All IPS processing, including traffic accepted by IPv4 and IPv6 traffic policies and IPv4 and IPv6 DoS policies is accelerated by the built-in SP3 processors.
- The FMC-XG2 is an FMC module with two 10Gb/s SPF+ interfaces that can be used on FortiGate-3950B and FortiGate-3951B units.
- The FortiGate-3140B also contains a built-in XG2 using ports 19 and 20.
- The ADM-XE2 is a dual-width AMC module with two 10Gb/s interfaces that can be used on FortiGate-3810A and FortiGate-5001A-DW systems.
- The ADM-FE8 is a dual-width AMC module with eight 1Gb/s interfaces that can be used with the FortiGate-3810A.
- The ASM-CE4 is a single-width AMC module with four 10/100/1000 Mb/s interfaces that can be used on FortiGate-3016B and FortiGate-3810A units.



Traffic is blocked if you enable IPS for traffic passing over inter-VDOM links if that traffic is being offloaded by an SP processor. If you disable SP offloading traffic will be allowed to flow. You can disable offloading in individual firewall policies by disabling `auto-asic-offload` for individual policies. You can also use the following command to disable all IPS offloading

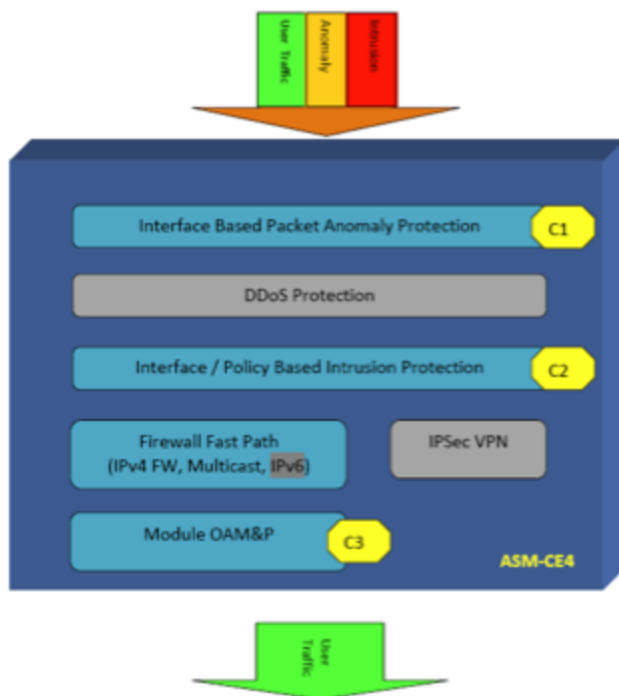
```
config ips global
    set np-accel-mode {none | basic}
    set cp-accel-mode {none | basic | advanced}
end
```

SP Processing Flow

SP processors provide an integrated high performance fast path multilayer solution for both intrusion protection and firewall functions. The multilayered protection starts from anomaly checking at packet level to ensure each packet is sound and reasonable. Immediately after that, a sophisticated set of interface based packet anomaly protection, DDoS protection, policy based intrusion protection, firewall fast path, and behavior based methods are employed to prevent DDoS attacks from the rest of system.

Then the packets enter an interface/policy based intrusion protection system,

where each packet is evaluated against a set of signatures. The end result is streams of user packets that are free of anomaly and attacks, entering the fast path system for unicast or multicast fast path forwarding.

SP processing flow

Displaying information about security processing modules

You can display information about installed SP modules using the CLI command

```
diagnose npu spm
```

For example, for the FortiGate-5101C:

```
FG-5101C # diagnose npu spm list
Available SP Modules:

ID Model      Slot      Interface
0 xh0        built-in  port1, port2, port3, port4,
                                     base1, base2, fabric1, fabric2
                                     eth10, eth11, eth12, eth13
                                     eth14, eth15, eth16, eth17
                                     eth18, eth19
```

You can also use this command to get more info about SP processing. This example shows how to display details about how the module is processing sessions using the syn proxy.

```
diagnose npu spm dos synproxy <sp_id>
```

This is a partial output of the command:

```
Number of proxied TCP connections :          0
Number of working proxied TCP connections :    0
Number of retired TCP connections :          0
Number of valid TCP connections :          0
Number of attacks, no ACK from client :       0
Number of no SYN-ACK from server :           0
Number of reset by server (service not supported): 0
Number of established session timeout :      0
Client timeout setting :                   3 Seconds
Server timeout setting :                   3 Seconds
```

Network processors (NP1, NP2, NP3, NP4 and NP6)

FortiASIC network processors work at the interface level to accelerate traffic by offloading traffic from the main CPU. Current models contain NP4 and NP6 network processors. Older FortiGate models include NP1 network processors (also known as FortiAccel, or FA2) and NP2 network processors.

The traffic that can be offloaded, maximum throughput, and number of network interfaces supported by each varies by processor model:

- NP6 supports offloading of most IPv4 and IPv6 traffic, IPsec VPN encryption, CAPWAP traffic, and multicast traffic. The NP6 has a capacity of 40 Gbps through 4 x 10 Gbps interfaces or 3 x 10 Gbps and 16 x 1 Gbps interfaces. For details about the NP6 processor, see [NP6 Acceleration on page 22](#) and for information about FortiGate models with NP6 processors, see [FortiGate NP6 architectures on page 33](#).
- NP4 supports offloading of most IPv4 firewall traffic and IPsec VPN encryption. The NP4 has a capacity of 20 Gbps through 2 x 10 Gbps interfaces. For details about NP4 processors, see [NP4 Acceleration on page 58](#) and for information about FortiGate models with NP4 processors, see [FortiGate NP4 architectures on page 73](#).
- NP2 supports IPv4 firewall and IPsec VPN acceleration. The NP2 has a capacity of 2 Gbps through 2 x 10 Gbps interfaces or 4 x 1 Gbps interfaces.

- NP1 supports IPv4 firewall and IPsec VPN acceleration with 2 Gbps capacity. The NP1 has a capacity of 2 Gbps through 2 x 1 Gbps interfaces.
 - The NP1 does not support frames greater than 1500 bytes. If your network uses jumbo frames, you may need to adjust the MTU (Maximum Transmission Unit) of devices connected to NP1 ports. Maximum frame size for NP2, NP4, and NP6 processors is 9000 bytes.
 - For both NP1 and NP2 network processors, ports attached to a network processor cannot be used for firmware installation by TFTP.



Sessions that require proxy-based and flow based security features (for example, virus scanning, IPS, application control and so on) are not fast pathed and must be processed by the CPU

Determining the network processors installed on your FortiGate unit

Use the following command to list the NP6 processors in your FortiGate unit:

```
diagnose npu np6 port-list
```

To list other network processors on your FortiGate unit, use the following CLI command.

```
get hardware npu <model> list
```

<model> can be legacy, np1, np2 or np4.

The output lists the interfaces that have the specified processor. For example, for a FortiGate-5001B:

```
get hardware npu np4 list
ID      Model      Slot      Interface
0       On-board      port1 port2 port3 port4
        fabric1 base1 npu0-vlink0 npu0-vlink1
1       On-board      port5 port6 port7 port8
        fabric2 base2 npu1-vlink0 npu1-vlink1
```

The npu0-vlink0, npu1-vlink1 etc interfaces are used for accelerating inter-VDOM links.

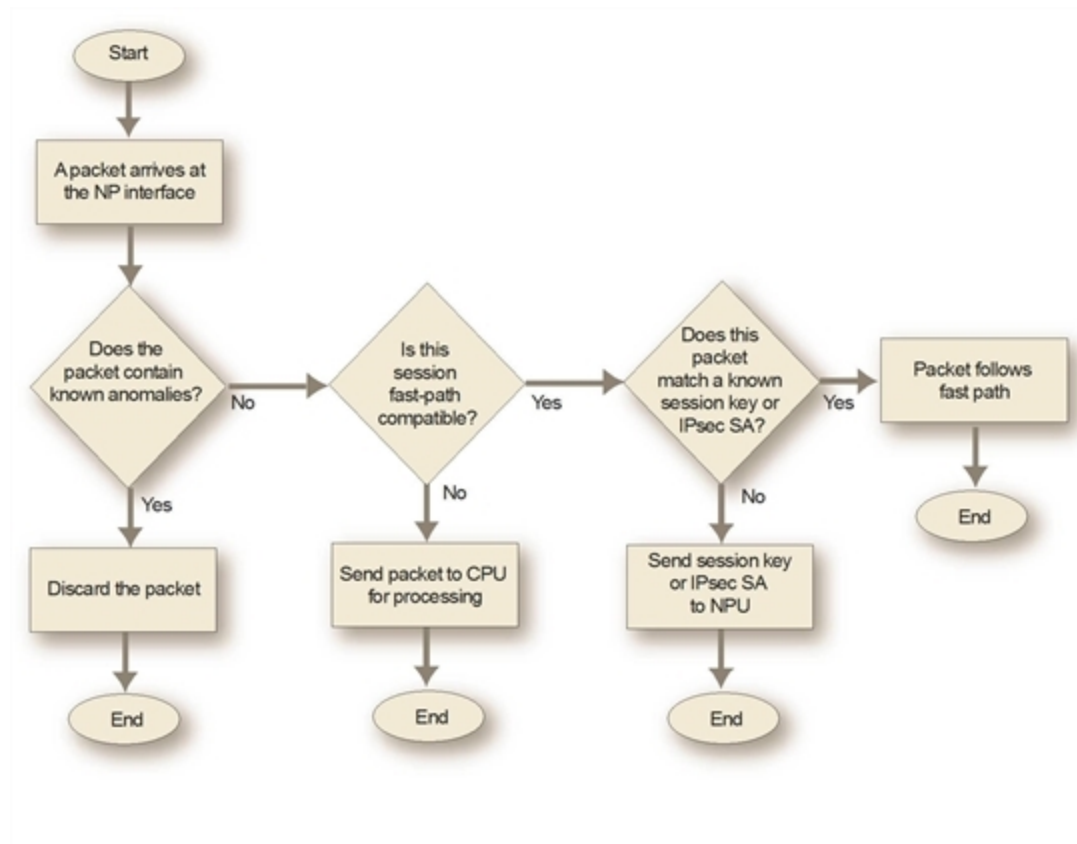
How NP hardware acceleration alters packet flow

NP hardware acceleration generally alters packet flow as follows:

- Packets initiating a session pass to the FortiGate unit's main processing resources (CPU).
- The FortiGate unit assesses whether the session matches fast path (offload) requirements. To be suitable for offloading, traffic must possess only characteristics that can be processed by the fast path. The list of requirements depends on the processor, see [NP6 session fast path requirements on page 22](#) or [NP4 session fast path requirements on page 59](#). If the session can be fast pathed, the FortiGate unit sends the session key or IPsec security association (SA) and configured firewall processing action to the appropriate network processor.
- Network processors continuously match packets arriving on their attached ports against the session keys and SAs they have received.
 - If a network processor's network interface is configured to perform hardware accelerated anomaly checks, the network processor drops or accepts packets that match the configured anomaly patterns. These checks are separate from and in advance of anomaly checks performed by IPS, which is not compatible with network processor offloading. See [Offloading NP4 anomaly detection on page 70](#).

- The network processor next checks for a matching session key or SA. If a matching session key or SA is found, and if the packet meets packet requirements, the network processor processes the packet according to the configured action and then sends the resulting packet. This is the actual offloading step. Performing this processing on the NP processor improves overall performance because the NP processor is optimized for this task. As well, overall FortiGate performance is improved because the CPU has fewer sessions to process.

NP network processor packet flow



- If a matching session key or SA is not found, or if the packet does not meet packet requirements, the packet cannot be offloaded. The network processor sends the data to the FortiGate unit's CPU, which processes the packet.

Encryption and decryption of IPsec traffic originating from the FortiGate can utilize network processor encryption capabilities.

Packet forwarding rates vary by the percentage of offloadable processing and the type of network processing required by your configuration, but are independent of frame size. For optimal traffic types, network throughput can equal wire speed.

NP processors and traffic logging and monitoring

Except for the NP6, network processors do not count offloaded packets, and offloaded packets are not logged by traffic logging and are not included in traffic statistics and traffic log reports.

NP6 processors support per-session traffic and byte counters, Ethernet MIB matching, and reporting through messages resulting in traffic statistics and traffic log reporting.

NP session offloading in HA active-active configuration

Network processors can improve network performance in active-active (load balancing) high availability (HA) configurations, even though traffic deviates from general offloading patterns, involving more than one network processor, each in a separate FortiGate unit. No additional offloading requirements apply.

Once the primary FortiGate unit's main processing resources send a session key to its network processor(s), network processor(s) on the primary unit can redirect any subsequent session traffic to other cluster members, reducing traffic redirection load on the primary unit's main processing resources.

As subordinate units receive redirected traffic, each network processor in the cluster assesses and processes session offloading independently from the primary unit. Session key states of each network processor are not part of synchronization traffic between HA members.

Configuring NP HMAC check offloading

Hash-based Message Authentication Code (HMAC) checks offloaded to network processors by default. You can enter the following command to disable this feature:

```
configure system global
    set ipsec-hmac-offload disable
end
```

Software switch interfaces and NP processors

FortiOS supports creating a software switch by grouping two or more FortiGate physical interfaces into a single virtual or software switch interface. All of the interfaces in this virtual switch act like interfaces in a hardware switch in that they all have the same IP address and can be connected to the same network. You create a software switch interface from the CLI using the command `config system switch-interface`.

The software switch is a bridge group of several interfaces, and the FortiGate CPU maintains the mac-port table for this bridge. As a result of this CPU involvement, traffic processed by a software switch interface is not offloaded to network processors.

Configuring NP accelerated IPsec VPN encryption/decryption offloading

Network processing unit (npu) settings configure offloading behavior for IPsec VPN. Configured behavior applies to all network processors in the FortiGate unit.

```
config system npu
    set enc-offload-antireplay {enable | disable}
    set dec-offload-antireplay {enable | disable}
    set offload-ipsec-host {enable | disable}
end
```


Variables	Description	Default
enc-offload-antireplay {enable disable}	Enable or disable offloading of IPsec encryption. This option is used only when replay detection is enabled in Phase 2 configuration. If replay detection is disabled, encryption is always offloaded.	disable
dec-offload-antireplay {enable disable}	Enable or disable offloading of IPsec decryption. This option is used only when replay detection is enabled in Phase 2 configuration. If replay detection is disabled, decryption is always offloaded.	enable
offload-ipsec-host {enable disable}	Enable or disable offloading of IPsec encryption of traffic from local host (FortiGate unit). Note: For this option to take effect, the FortiGate unit must have previously sent the security association (SA) to the network processor.	disable

Example

You could configure the offloading of encryption and decryption for an IPsec SA that was sent to the network processor.

```
config system npu
    set enc-offload-antireplay enable
    set dec-offload-antireplay enable
    set offload-ipsec-host enable
end
```

Disabling NP acceleration for individual IPsec VPN phase 1s

Use the following command to disable NP offloading for an interface-based IPsec VPN phase 1:

```
config vpn ipsec phase1-interface
    edit phase-1-name
        set npu-offload disable
    end
```

Use the following command to disable NP offloading for a policy-based IPsec VPN phase 1:

```
config vpn ipsec phase1
    edit phase-1-name
        set npu-offload disable
    end
```

The `npu-offload` option is enabled by default.

Disabling NP offloading for unsupported IPsec encryption or authentication algorithms

In general, more recent IPsec VPN encryption and authentication algorithms may not be supported by older NP processors. For example, NP4 network processors do not support SHA-256, SHA-384, and SHA-512. IPsec traffic with unsupported algorithms is not offloaded and instead is processed by the FortiGate CPU. In addition, this

configuration may cause packet loss and other performance issues. If you experience packet loss or performance problems you should set the `npu-offload` option to `disable`. Future FortiOS versions should prevent selecting algorithms not supported by the hardware.

Disabling NP offloading for firewall policies

Use the following options to disable NP offloading for specific security policies:

For IPv4 security policies.

```
config firewall policy
edit 1
set auto-asic-offload disable
end
```

For IPv6 security policies.

```
config firewall policy6
edit 1
set auto-asic-offload disable
end
```

For multicast security policies.

```
config firewall multicast-policy
edit 1
set auto-asic-offload disable
end
```

Enabling strict protocol header checking disables all hardware acceleration

You can use the following command to cause the FortiGate to apply strict header checking to verify that a packet is part of a session that should be processed. Strict header checking includes verifying the layer-4 protocol header length, the IP header length, the IP version, the IP checksum, IP options, and verifying that ESP packets have the correct sequence number, SPI, and data length. If the packet fails header checking it is dropped by the FortiGate unit.

```
config system global
check-protocol-header strict
end.
```

Enabling strict header checking disables all hardware acceleration. This includes NP, SP, and CP processing.

sFlow and NetFlow and hardware acceleration

Configuring sFlow or NetFlow on any FortiGate interface disables all NP4 and NP6 offloading for all traffic.

Checking that traffic is offloaded by NP processors

A number of diagnose commands can be used to verify that traffic is being offloaded.

Using the packet sniffer

Use the packet sniffer to verify that traffic is offloaded. Offloaded traffic is not picked up by the packet sniffer so if you are sending traffic through the FortiGate unit and it is not showing up on the packet sniffer you can conclude that it is offloaded.

```
diag sniffer packet port1 <option>
```

Checking the firewall session offload tag

Use the `diagnose sys session list` command to display sessions. If the output for a session includes the `npu info` field you should see information about session being offloaded. If the output doesn't contain an `npu info` field then the session has not been offloaded.

```
diagnose sys session list
session info: proto=6 proto_state=01 duration=34 expire=3565 timeout=3600
             flags=00000000 sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/
state=may_dirty npu
statistic(bytes/packets/allow_err): org=295/3/1 reply=60/1/1 tuples=2
orgin->sink: org pre->post, reply pre->post dev=48->6/6->48 gwy=10.1.100.11/11.11.11.1
hook=pre dir=org act=noop 172.16.200.55:56453->10.1.100.11:80(0.0.0.0:0)
hook=post dir=reply act=noop 10.1.100.11:80->172.16.200.55:56453(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 id_policy_id=0 auth_info=0 chk_client_info=0 vd=4
serial=0000091c tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
per_ip_bandwidth meter: addr=172.16.200.55, bps=393
npu_state=00000000
npu info: flag=0x81/0x81, offload=4/4, ips_offload=0/0, epid=1/23, ipid=23/1,
vlan=32779/0
```

Verifying IPsec VPN traffic offloading

The following commands can be used to verify IPsec VPN traffic offloading to NP processors.

```
diagnose vpn ipsec status
NP1/NP2/NP4_0/sp_0_0:
  null: 0 0
  des: 0 0
  3des: 4075 4074
  aes: 0 0
  aria: 0 0
  seed: 0 0
  null: 0 0
  md5: 4075 4074
  sha1: 0 0
  sha256: 0 0
  sha384: 0 0
  sha512: 0 0

diagnose vpn tunnel list
```

```
list all ipsec tunnel in vd 3
-----
name=p1-vdom1 ver=1 serial=5 11.11.11.1:0->11.11.11.2:0 lgwy=static tun=tunnel
mode=auto bound_if=47
proxyid_num=1 child_num=0 refcnt=8 ilast=2 olast=2
stat: rxp=3076 txp=1667 rxb=4299623276 txb=66323
dpd: mode=active on=1 idle=5000ms retry=3 count=0 seqno=20
natt: mode=none draft=0 interval=0 remote_port=0
proxyid=p2-vdom1 proto=0 sa=1 ref=2 auto_negotiate=0 serial=1
src: 0:0.0.0.0/0.0.0.0:0
dst: 0:0.0.0.0/0.0.0.0:0
SA: ref=6 options=0000000e type=00 soft=0 mtu=1436 expire=1736 replaywin=2048 seqno=680
life: type=01 bytes=0/0 timeout=1748/1800
dec: spi=ae01010c esp=3des key=24 18e021bcace225347459189f292fbc2e4677563b07498a07
ah=md5 key=16 b4f44368741632b4e33e5f5b794253d3
enc: spi=ae01010d esp=3des key=24 42c94a8a2f72a44f9a3777f8e6aa3b24160b8af15f54a573
ah=md5 key=16 6214155f76b63a93345dcc9ec02d6415
dec:pkts/bytes=3073/4299621477, enc:pkts/bytes=1667/66375
  npu_flag=03 npu_rgwy=11.11.11.2 npu_lgwy=11.11.11.1 npu_selid=4

diagnose sys session list
session info: proto=6 proto_state=01 duration=34 expire=3565 timeout=3600
  flags=00000000 sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/p1-vdom2
state=re may_dirty npu
statistic(bytes/packets/allow_err): org=112/2/1 reply=112/2/1 tuples=2
orgin->sink: org pre->post, reply pre->post dev=57->7/7->57 gw=10.1.100.11/11.11.11.1
hook=pre dir=org act=noop 172.16.200.55:35254->10.1.100.11:80(0.0.0.0:0)
hook=post dir=reply act=noop 10.1.100.11:80->172.16.200.55:35254(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 id_policy_id=0 auth_info=0 chk_client_info=0 vd=4
serial=00002d29 tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
per_ip_bandwidth meter: addr=172.16.200.55, bps=260
npu_state=00000000
npu info: flag=0x81/0x82, offload=7/7, ips_offload=0/0, epid=1/3, ipid=3/1,
  vlan=32779/0
```

Controlling IPS NPx and CPx acceleration

You can use the following commands to enable or disable acceleration of IPS processing by NPx and CPx processors:

```
config ips global
  set np-accel-mode {none | basic}
  set cp-accel-mode {none | basic | advanced}
end
```

The network processor (NP) acceleration modes are:

none: Network Processor acceleration disabled

basic: Basic Network Processor acceleration enabled

The content processor (CP) acceleration modes are:

- `none`: Content Processor acceleration disabled
- `basic`: Basic Content Processor acceleration enabled
- `advanced`: Advanced Content Processor acceleration enabled

Dedicated Management CPU

The web-based manager and CLI of FortiGate units with NP6 and NP4 processors may become unresponsive when the system is under heavy processing load because NP6 or NP4 interrupts overload the CPUs preventing CPU cycles from being used for management tasks. You can resolve this issue by using the following command to dedicate CPU core 0 to management tasks.

```
config system npu
    set dedicated-management-cpu {enable | disable}
end
```

All management tasks are then processed by CPU 0 and NP6 or NP4 interrupts are handled by the remaining CPU cores.

NP6 Acceleration

NP6 network processors provide fastpath acceleration by offloading communication sessions from the FortiGate CPU. When the first packet of a new session is received by an interface connected to an NP6 processor, just like any session connecting with any FortiGate interface, the session is forwarded to the FortiGate CPU where it is matched with a security policy. If the session is accepted by a security policy and if the session can be offloaded its session key is copied to the NP6 processor that received the packet. All of the rest of the packets in the session are intercepted by the NP6 processor and fast-pathed out of the FortiGate unit to their destination without ever passing through the FortiGate CPU. The result is enhanced network performance provided by the NP6 processor plus the network processing load is removed from the CPU. In addition the NP6 processor can handle some CPU intensive tasks, like IPsec VPN encryption/decryption.

Session keys (and IPsec SA keys) are stored in the memory of the NP6 processor that is connected to the interface that received the packet that started the session. All sessions are fast-pathed and accelerated, even if they exit the FortiGate unit through an interface connected to another NP6. There is no dependence on getting the right pair of interfaces since the offloading is done by the receiving NP6.

The key to making this possible is an Integrated Switch Fabric (ISF) that connects the NP6s and the FortiGate unit interfaces together. Many FortiGate units with NP6 processors also have an ISF. The ISF allows any port connectivity. All ports and NP6s can communicate with each other over the ISF. There are no special ingress and egress fast path requirements as long as traffic enters and exits on interfaces connected to the same ISF.

Some FortiGate units, such as the FortiGate-1000D include multiple NP6 processors that are not connected by an ISF. Because the ISF is not present fast path acceleration is supported only between interfaces connected to the same NP6 processor. Since the ISF introduces some latency, models with no ISF provide low-latency network acceleration between network interfaces connected to the same NP6 processor.

There are at least two limitations to keep in mind:

- The capacity of each NP6 processor. An individual NP6 processor can support between 10 and 16 million sessions. This number is limited by the amount of memory the processor has. Once an NP6 processor hits its session limit, sessions that are over the limit are sent to the CPU. You can avoid this problem by as much as possible distributing incoming sessions evenly among the NP6 processors. To be able to do this you need to be aware of which interfaces connect to which NP6 processors and distribute incoming traffic accordingly.
- Some FortiGate units may use some NP6 processors for special functions. For example, ports 25 to 32 of the FortiGate-3700D can be used for low latency offloading.

NP6 session fast path requirements

NP6 processors can offload the following traffic and services:

- IPv4 and IPv6 traffic and NAT64 and NAT46 traffic (as well as IPv4 and IPv6 versions of the following traffic types where appropriate)
- Link aggregation (LAG) (IEEE 802.3ad) traffic (see [Increasing NP6 offloading capacity using link aggregation groups \(LAGs\) on page 24](#))
- TCP, UDP, ICMP and SCTP traffic
- IPsec VPN traffic, and offloading of IPsec encryption/decryption (including SHA2-256 and SHA2-512)

- Anomaly-based intrusion prevention, checksum offload and packet defragmentation
- SIT and IPv6 Tunnelling sessions
- Multicast traffic (including Multicast over IPsec)
- CAPWAP and wireless bridge traffic tunnel encapsulation to enable line rate wireless forwarding from FortiAP devices
- Traffic shaping and priority queuing for both shared and per IP traffic shaping. An NP6 processor has 16 million queues for traffic shaping and statistics counting.
- Syn proxying
- Inter-VDOM link traffic

Sessions that are offloaded must be fast path ready. For a session to be fast path ready it must meet the following criteria:

- Layer 2 type/length must be 0x0800 for IPv4 or 0x86dd for IPv6 (IEEE 802.1q VLAN specification is supported)
- Layer 3 protocol can be IPv4 or IPv6
- Layer 4 protocol can be UDP, TCP, ICMP, or SCTP
- In most cases, Layer 3 / Layer 4 header or content modification sessions that require a session helper can be offloaded.
- Local host traffic (originated by the FortiGate unit) can be offloaded
- Application layer content modification is not supported (the firewall policy that accepts the session must not include virus scanning, web filtering, DLP, application control, IPS, email filtering, SSL/SSH inspection, VoIP or ICAP)



If you disable anomaly checks by Intrusion Prevention (IPS), you can still enable hardware accelerated anomaly checks using the `fp-anomaly` field of the `config system interface` CLI command. See [Configuring individual NP6 processors on page 28](#).

If a session or is not fast path ready, the FortiGate unit will not send the session key or IPsec SA key to the NP6 processor. Without the session key, all session key lookup by a network processor for incoming packets of that session fails, causing all session packets to be sent to the FortiGate unit's main processing resources, and processed at normal speeds.

If a session is fast path ready, the FortiGate unit will send the session key or IPsec SA key to the network processor. Session key or IPsec SA key lookups then succeed for subsequent packets from the known session or IPsec SA.

Packet fast path requirements

Packets within the session must then also meet packet requirements.

- Incoming packets must not be fragmented.
- Outgoing packets must not require fragmentation to a size less than 385 bytes. Because of this requirement, the configured MTU (Maximum Transmission Unit) for a network processor's network interfaces must also meet or exceed the network processors' supported minimum MTU of 385 bytes.

Mixing fast path and non-fast path traffic

If packet requirements are not met, an individual packet will be processed by the FortiGate CPU regardless of whether other packets in the session are offloaded to the NP6.

Also, in some cases, a protocol's session(s) may receive a mixture of offloaded and non-offloaded processing. For example, VoIP control packets may not be offloaded but VoIP data packets (voice packets) may be offloaded.

Viewing your FortiGate NP6 processor configuration

Use the following command to view the NP6 processor configuration of your FortiGate unit:

```
diagnose npu np6 port-list
```

For example output of this command for different FortiGate models, see [FortiGate NP6 architectures on page 33](#).

Increasing NP6 offloading capacity using link aggregation groups (LAGs)

NP6 processors can offload sessions received by interfaces in link aggregation groups (LAGs) (IEEE 802.3ad). A LAG combines more than one physical interface into a group that functions like a single interface with a higher capacity than a single physical interface. For example, you could use a LAG if you want to offload sessions on a 30 Gbps link by adding three 10-Gbps interfaces to the same LAG.

All offloaded traffic types are supported by LAGs, including IPsec VPN traffic. Just like with normal interfaces, traffic accepted by a LAG is offloaded by the NP6 processor connected to the interfaces in the LAG that receive the traffic to be offloaded. If all interfaces in a LAG are connected to the same NP6 processor, traffic received by that LAG is offloaded by that NP6 processor. The amount of traffic that can be offloaded is limited by the capacity of the NP6 processor.

If a FortiGate has two or more NP6 processors connected by an integrated switch fabric (ISF), you can use LAGs to increase offloading by sharing the traffic load across multiple NP6 processors. You do this by adding physical interfaces connected to different NP6 processors to the same LAG.

Adding a second NP6 processor to a LAG effectively doubles the offloading capacity of the LAG. Adding a third further increases offloading. The actual increase in offloading capacity may not actually be doubled by adding a second NP6 or tripled by adding a third. Traffic and load conditions and other factors may limit the actual offloading result.

The increase in offloading capacity offered by LAGs and multiple NP6s is supported by the integrated switch fabric (ISF) that allows multiple NP6 processors to share session information. Most FortiGate units with multiple NP6 processors also have an ISF. However, the FortiGate-1000D does not have an ISF. On this model and others that have more than one NP6 and no ISF, if you attempt to add interfaces connected to different NP6 processors to a LAG the system displays an error message.

There are also a few limitations to LAG NP6 offloading support for IPsec VPN:

- IPsec VPN anti-replay protection cannot be used if IPsec is configured on a LAG that has interfaces connected to multiple NP6 processors.
- Because the encrypted traffic for one IPsec VPN tunnel has the same 5-tuple, the traffic from one tunnel can only be balanced to one interface in a LAG. This limits the maximum throughput for one IPsec VPN tunnel in an NP6 LAG group to 10Gbps.

Configuring Inter-VDOM link acceleration with NP6 processors

FortiGate units with NP6 processors include inter-VDOM links that can be used to accelerate inter-VDOM link traffic.

- For a FortiGate unit with two NP6 processors there are two accelerated inter-VDOM links, each with two interfaces:
 - **npu0_vlink:**
 - npu0_vlink0
 - npu0_vlink1
 - **npu1_vlink:**
 - npu1_vlink0
 - npu1_vlink1

These interfaces are visible from the GUI and CLI. For a FortiGate unit with NP6 interfaces, enter the following CLI command to display the NP6-accelerated inter-VDOM links:

```
get system interface
...
== [ npu0_vlink0 ]
name: npu0_vlink0 mode: static ip: 0.0.0.0 0.0.0.0 status: down netbios-forward: disable
type: physical sflow-sampler: disable explicit-web-proxy: disable explicit-ftp-proxy:
disable mtu-override: disable wccp: disable drop-overlapped-fragment: disable drop-
fragment: disable

== [ npu0_vlink1 ]
name: npu0_vlink1 mode: static ip: 0.0.0.0 0.0.0.0 status: down netbios-forward: disable
type: physical sflow-sampler: disable explicit-web-proxy: disable explicit-ftp-proxy:
disable mtu-override: disable wccp: disable drop-overlapped-fragment: disable drop-
fragment: disable

== [ npu1_vlink0 ]
name: npu1_vlink0 mode: static ip: 0.0.0.0 0.0.0.0 status: down netbios-forward: disable
type: physical sflow-sampler: disable explicit-web-proxy: disable explicit-ftp-proxy:
disable mtu-override: disable wccp: disable drop-overlapped-fragment: disable drop-
fragment: disable

== [ npu1_vlink1 ]
name: npu1_vlink1 mode: static ip: 0.0.0.0 0.0.0.0 status: down netbios-forward: disable
type: physical sflow-sampler: disable explicit-web-proxy: disable explicit-ftp-proxy:
disable mtu-override: disable wccp: disable drop-overlapped-fragment: disable drop-
fragment: disable
...
```

By default the interfaces in each inter-VDOM link are assigned to the root VDOM. To use these interfaces to accelerate inter-VDOM link traffic, assign each interface in the pair to the VDOMs that you want to offload traffic between. For example, if you have added a VDOM named New-VDOM to a FortiGate unit with NP4 processors, you can go to **System > Network > Interfaces** and edit the **npu0-vlink1** interface and set the **Virtual Domain** to **New-VDOM**. This results in an accelerated inter-VDOM link between root and New-VDOM. You can also do this from the CLI:

```
config system interface
  edit npu0-vlink1
    set vdom New-VDOM
  end
```

Using VLANs to add more accelerated Inter-VDOM links

You can add VLAN interfaces to the accelerated inter-VDOM links to create inter-VDOM links between more VDOMs. For the links to work, the VLAN interfaces must be added to the same inter-VDOM link, must be on the same subnet, and must have the same VLAN ID.

For example, to accelerate inter-VDOM link traffic between VDOMs named Marketing and Engineering using VLANs with VLAN ID 100 go to **System > Network > Interfaces** and select **Create New** to create the VLAN interface associated with the Marketing VDOM:

Name	Marketing-link
Type	VLAN
Interface	npu0_vlink0
VLAN ID	100
Virtual Domain	Marketing
IP/Network Mask	172.20.120.12/24

Create the inter-VDOM link associated with Engineering VDOM:

Name	Engineering-link
Type	VLAN
Interface	npu0_vlink1
VLAN ID	100
Virtual Domain	Engineering
IP/Network Mask	172.20.120.22/24

Or do the same from the CLI:

```
config system interface
  edit Marketing-link
    set vdom Marketing
    set ip 172.20.120.12/24
    set interface npu0_vlink0
    set vlanid 100
  next
  edit Engineering-link
    set vdom Engineering
    set ip 172.20.120.22/24
    set interface npu0_vlink1
    set vlanid 100
```

Confirm that the traffic is accelerated

Use the following CLI commands to obtain the interface index and then correlate them with the session entries. In the following example traffic was flowing between new accelerated inter-VDOM links and physical ports port1 and port 2 also attached to the NP6 processor.

```
diagnose ip address list
IP=172.31.17.76->172.31.17.76/255.255.252.0 index=5 devname=port1
IP=10.74.1.76->10.74.1.76/255.255.252.0 index=6 devname=port2
IP=172.20.120.12->172.20.120.12/255.255.255.0 index=55 devname=IVL-VLAN1_ROOT
IP=172.20.120.22->172.20.120.22/255.255.255.0 index=56 devname=IVL-VLAN1_VDOM1

diagnose sys session list
session info: proto=1 proto_state=00 duration=282 expire=24 timeout=0 session info:
    proto=1 proto_state=00 duration=124 expire=59 timeout=0 flags=00000000
    sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/
state=may_dirty npu
statistic(bytes/packets/allow_err): org=180/3/1 reply=120/2/1 tuples=2
origin->sink: org pre->post, reply pre->post dev=55->5/5->55
    gwy=172.31.19.254/172.20.120.22
hook=post dir=org act=snat 10.74.2.87:768->10.2.2.2:8(172.31.17.76:62464)
hook=pre dir=reply act=dnat 10.2.2.2:62464->172.31.17.76:0(10.74.2.87:768)
misc=0 policy_id=4 id_policy_id=0 auth_info=0 chk_client_info=0 vd=0
serial=0000004e tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
per_ip_bandwidth meter: addr=10.74.2.87, bps=880
npu_state=00000000
npu info: flag=0x81/0x81, offload=8/8, ips_offload=0/0, epid=160/218, ipid=218/160,
vlan=32769/0

session info: proto=1 proto_state=00 duration=124 expire=20 timeout=0 flags=00000000
    sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/
state=may_dirty npu
statistic(bytes/packets/allow_err): org=180/3/1 reply=120/2/1 tuples=2
origin->sink: org pre->post, reply pre->post dev=6->56/56->6
    gwy=172.20.120.12/10.74.2.87
hook=pre dir=org act=noop 10.74.2.87:768->10.2.2.2:8(0.0.0.0:0)
hook=post dir=reply act=noop 10.2.2.2:768->10.74.2.87:0(0.0.0.0:0)
misc=0 policy_id=3 id_policy_id=0 auth_info=0 chk_client_info=0 vd=1
serial=0000004d tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
per_ip_bandwidth meter: addr=10.74.2.87, bps=880
npu_state=00000000
npu info: flag=0x81/0x81, offload=8/8, ips_offload=0/0, epid=219/161, ipid=161/219,
vlan=0/32769
total session 2
```

Configuring individual NP6 processors

You can use the `config system np6` command to configure a wide range of settings for the NP6 processors in your FortiGate unit including enabling/disabling fastpath and low latency and adjusting session timeouts. As well you can set anomaly checking for IPv4 and IPv6 traffic. You can also configure different settings for each NP6 processor.

The settings that you configure for an NP6 processor with the `config system np6` command apply to traffic processed by all interfaces connected to the NP6 processor. This includes the physical interfaces connected to the NP6 processor as well as all subinterfaces, VLAN interfaces, IPsec interfaces, LAGs and so on associated with the physical interfaces connected to the NP6 processor.



Some of the options for this command apply anomaly checking for NP6 sessions in the same way as the command described in [Offloading NP4 anomaly detection on page 70](#) applies anomaly checking for for NP4 sessions.

```
config system np6
  edit <np6-processor-name>
    set fastpath {disable | enable}
    set low-latency-mode {disable | enable}
    set per-session-accounting {disable | enable}
    set session-timeout-random-range <range>
    set garbage-session-collector {disable | enable}
    set session-collector-interval <range>
    set session-timeout-interval <range>
    set session-timeout-random-range <range>
    set session-timeout-fixed {disable | enable}
    config fp-anomaly-v4
      set icmp-frag {allow | drop | trap-to-host}
      set icmp-land {allow | drop | trap-to-host}
      set ipv4-land {allow | drop | trap-to-host}
      set ipv4-optlsrr {allow | drop | trap-to-host}
      set ipv4-optrrr {allow | drop | trap-to-host}
      set ipv4-optsecurity {allow | drop | trap-to-host}
      set ipv4-optssrr {allow | drop | trap-to-host}
      set ipv4-optstream {allow | drop | trap-to-host}
      set ipv4-opttimestamp {allow | drop | trap-to-host}
      set ipv4-proto-err {allow | drop | trap-to-host}
      set ipv4-unknopt {allow | drop | trap-to-host}
      set tcp-land {allow | drop | trap-to-host}
      set tcp-syn-fin {allow | drop | trap-to-host}
      set tcp-winnuke {allow | drop | trap-to-host}
      set tcp_fin_noack {allow | drop | trap-to-host}
      set tcp_fin_only {allow | drop | trap-to-host}
      set tcp_no_flag {allow | drop | trap-to-host}
      set tcp_syn_data {allow | drop | trap-to-host}
      set udp-land {allow | drop | trap-to-host}
    end
    config fp-anomaly-v6
      set ipv6-daddr_err {allow | drop | trap-to-host}
      set ipv6-land {allow | drop | trap-to-host}
```

```

set ipv6-optendpid {allow | drop | trap-to-host}
set ipv6-opthomeaddr {allow | drop | trap-to-host}
set ipv6-optinvld {allow | drop | trap-to-host}
set ipv6-optjumbo {allow | drop | trap-to-host}
set ipv6-optnsap {allow | drop | trap-to-host}
set ipv6-optralert {allow | drop | trap-to-host}
set ipv6-opttunnel {allow | drop | trap-to-host}
set ipv6-proto-err {allow | drop | trap-to-host}
set ipv6-saddr_err {allow | drop | trap-to-host}
set ipv6-unknopt {allow | drop | trap-to-host}
end

```

Command syntax

Command	Description	Default
fastpath {disable enable}	Enable fastpath acceleration to offload sessions to the NP6 processor. You can disable fastpath if you don't want the NP6 processor to offload sessions.	enable
low-latency-mode {disable enable}	Enable low-latency mode. In low latency mode the integrated switch fabric is bypassed. Low latency mode requires that packet enter and exit using the same NP6 processor. This option is only available for NP6 processors that can operate in low-latency mode.	disable
per-session-accounting {disable enable}	Record traffic log messages for offloaded sessions. Enabling this feature reduces performance..	disable
garbage-session-collector {disable enable}	Enable deleting expired or garbage sessions.	disable
session-collector-interval <range>	Set the expired or garbage session collector time interval in seconds. The range is 1 to 100 seconds.	8
session-timeout-interval <range>	Set the timeout for inactive sessions. The range is 0 to 1000 seconds.	40
session-timeout-random-range <range>	Set the random timeout for inactive sessions. The range is 0 to 1000 seconds.	8
session-timeout-fixed {disable enable}	Force session timeouts at fixed instead of random intervals.	disable
config fp-anomaly-v4		

Command	Description	Default
<code>fp-anomaly-v4</code>	Configure how the NP6 processor does IPv4 traffic anomaly protection. You can configure the NP6 processor to allow or drop the packets associated with an attack or forward the packets that are associated with the attack to FortiOS (called “trap-to-host”). Selecting “trap-to-host” turns off NP6 anomaly protection for that anomaly. If you require anomaly protection you can enable it with a DoS policy.	
<code>icmp-frag {allow drop trap-to-host}</code>	Detects Layer 3 fragmented packets that could be part of a layer 4 ICMP anomalies.	allow
<code>icmp-land {allow drop trap-to-host}</code>	Detects ICMP land anomalies.	trap-to-host
<code>ipv4-land {allow drop trap-to-host}</code>	Detects IPv4 land anomalies.	trap-to-host
<code>ipv4-optlsrr {allow drop trap-to-host}</code>	Detects IPv4 with loose source record route option anomalies.	trap-to-host
<code>ipv4-optrr {allow drop trap-to-host}</code>	Detects IPv4 with record route option anomalies.	trap-to-host
<code>ipv4-optsecurity {allow drop trap-to-host}</code>	Detects security option anomalies.	trap-to-host
<code>ipv4-optssrr {allow drop trap-to-host}</code>	Detects IPv4 with strict source record route option anomalies.	trap-to-host
<code>ipv4-optstream {allow drop trap-to-host}</code>	Detects stream option anomalies.	trap-to-host
<code>ipv4-opttimestamp {allow drop trap-to-host}</code>	Detects timestamp option anomalies.	trap-to-host
<code>ipv4-proto-err {allow drop trap-to-host}</code>	Detects invalid layer 4 protocol anomalies.	trap-to-host
<code>ipv4-unknopt {allow drop trap-to-host}</code>	Detects unknown option anomalies.	trap-to-host
<code>tcp-land {allow drop trap-to-host}</code>	Detects TCP land anomalies.	trap-to-host
<code>tcp-syn-fin {allow drop trap-to-host}</code>	Detects TCP SYN flood SYN/FIN flag set anomalies.	allow

Command	Description	Default
<code>tcp-winnuke {allow drop trap-to-host}</code>	Detects TCP WinNuke anomalies.	trap-to-host
<code>tcp_fin_noack {allow drop trap-to-host}</code>	Detects TCP SYN flood with FIN flag set without ACK setting anomalies.	trap-to-host
<code>tcp_fin_only {allow drop trap-to-host}</code>	Detects TCP SYN flood with only FIN flag set anomalies.	trap-to-host
<code>tcp_no_flag {allow drop trap-to-host}</code>	Detects TCP SYN flood with no flag set anomalies.	allow
<code>tcp_syn_data {allow drop trap-to-host}</code>	Detects TCP SYN flood packets with data anomalies.	allow
<code>udp-land {allow drop trap-to-host}</code>	Detects UDP land anomalies.	trap-to-host
config fp-anomaly-v6		
<code>fp-anomaly-v6</code>	Configure how the NP6 processor does IPv6 traffic anomaly protection. You can configure the NP6 processor to allow or drop the packets associated with an attack or forward the packets that are associated with the attack to FortiOS (called “trap-to-host”). Selecting “trap-to-host” turns off NP6 anomaly protection for that anomaly. If you require anomaly protection you can enable it with a DoS policy.	
<code>ipv6-daddr_err {allow drop trap-to-host}</code>	Detects destination address as unspecified or loopback address anomalies.	trap-to-host
<code>ipv6-land {allow drop trap-to-host}</code>	Detects IPv6 land anomalies	trap-to-host
<code>ipv6-optendpid {allow drop trap-to-host}</code>	Detects end point identification anomalies.	trap-to-host
<code>ipv6-opthomeaddr {allow drop trap-to-host}</code>	Detects home address option anomalies.	trap-to-host
<code>ipv6-optinvld {allow drop trap-to-host}</code>	Detects invalid option anomalies.	trap-to-host
<code>ipv6-optjumbo {allow drop trap-to-host}</code>	Detects jumbo options anomalies.	trap-to-host

Command	Description	Default
<code>ipv6-optnsap {allow drop trap-to-host}</code>	Detects network service access point address option anomalies.	trap-to-host
<code>ipv6-optralert {allow drop trap-to-host}</code>	Detects router alert option anomalies.	trap-to-host
<code>ipv6-opttunnel {allow drop trap-to-host}</code>	Detects tunnel encapsulation limit option anomalies.	trap-to-host
<code>ipv6-proto-err {allow drop trap-to-host}</code>	Detects layer 4 invalid protocol anomalies.	trap-to-host
<code>ipv6-saddr_err {allow drop trap-to-host}</code>	Detects source address as multicast anomalies.	trap-to-host
<code>ipv6-unknopt {allow drop trap-to-host}</code>	Detects unknown option anomalies.	trap-to-host

Enabling per-session accounting for offloaded NP6 sessions

By default FortiOS does not record log messages for offloaded NP6 and NP4 sessions. This also means that traffic monitoring does not report correct session counts, byte counts and packet counts for offloaded NP6 and NP4 sessions.

However, for NP6 processors, you can use the following command to enable per-session accounting for each NP6 processor in the FortiGate unit. Per session accounting is a logging feature that allows the FortiGate to report the correct bytes/pkt numbers per session for sessions offloaded to an NP6 processor. Per-session accounting is not enabled by default because it reduces NP6 offloading performance. So you should only enable per-session accounting if you need the accounting information.

Some FortiOS versions enable per-session-accounting automatically if you enable traffic logging in a firewall policy that allows traffic to pass through an interface attached to an NP6 processor.

Enabling per-session accounting only supports traffic log messages and does not provide traffic flow data for sFlow or NetFlow.

For example, to enable session accounting for the first and second NP6 processors (np6_0 and np6_1):

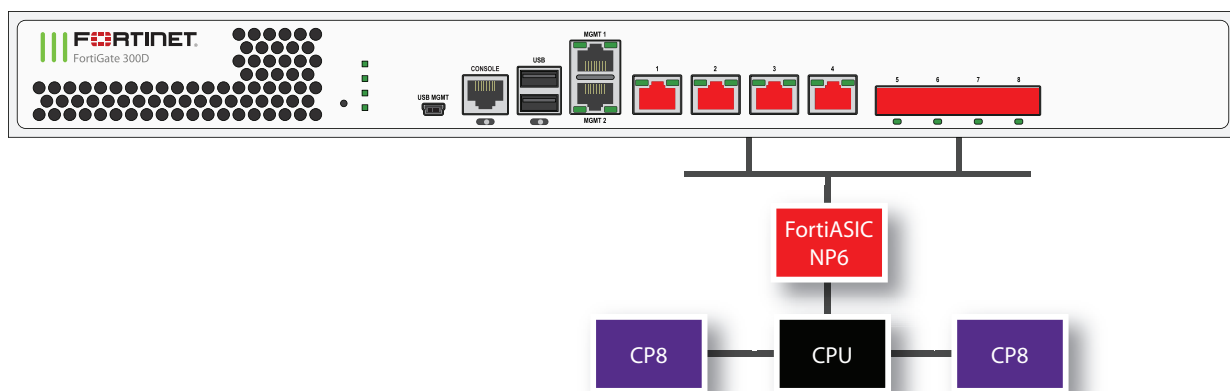
```
config system np6
  edit np6_0
    set per-session-accounting enable
  next
  edit np6_1
    set per-session-accounting enable
end
```


FortiGate NP6 architectures

Many FortiGate models can offload some types of network traffic processing from main processing resources to specialized network processors. If your network has a significant volume of traffic that is suitable for offloading, this hardware acceleration can significantly improve your network throughput.

FortiGate-300D fast path architecture

The FortiGate-300D includes one NP6 processor connected to four 1Gb RJ-45 Ethernet ports (port1-4) and four 1Gb SFP interfaces (port5-port8).

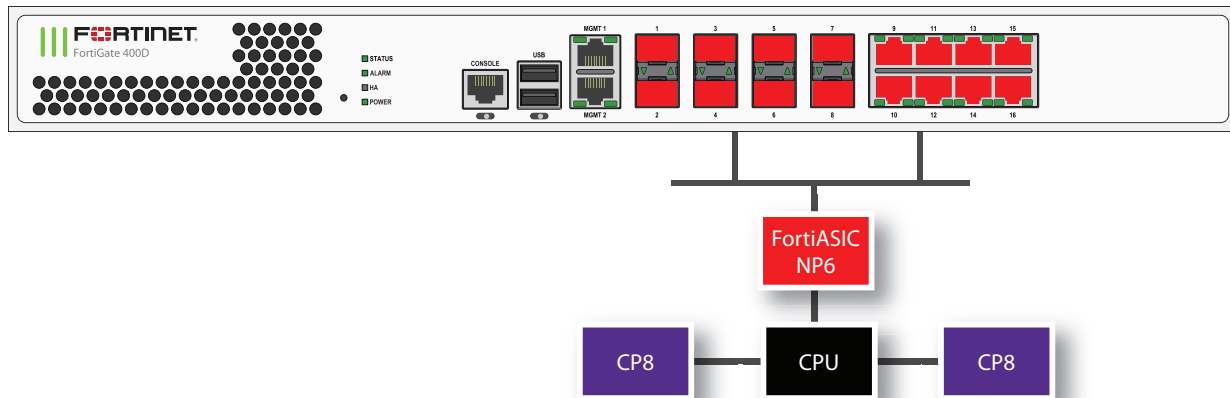


You can use the following command to display the FortiGate-300D NP6 configuration. The command output shows one NP6 named NP6_0 and the interfaces (ports) connected to it.

```
diagnose npu np6 port-list
Chip  XAUI Ports  Max  Cross-chip
      Speed offloading
-----
np6_0  0
      1  port5  1G  Yes
      1  port7  1G  Yes
      1  port8  1G  Yes
      1  port6  1G  Yes
      1  port3  1G  Yes
      1  port4  1G  Yes
      1  port1  1G  Yes
      1  port2  1G  Yes
      2
      3
-----
```

FortiGate-400D fast path architecture

The FortiGate-400D includes one NP6 processor connected to eight 1Gb SFP interfaces (port1-port8) and eight 1Gb RJ-45 Ethernet ports (port9-16).

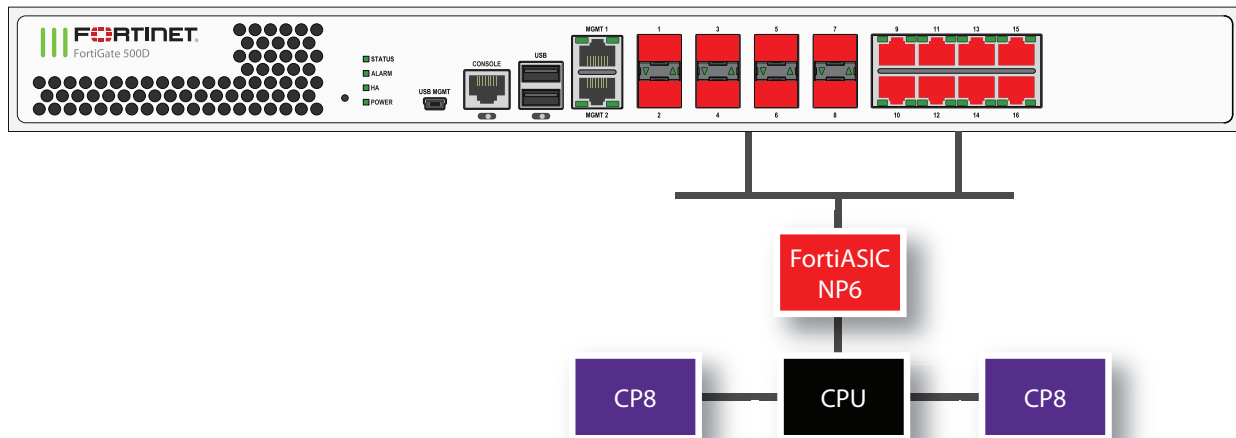


You can use the following command to display the FortiGate-400D NP6 configuration. The command output shows one NP6 named NP6_0 and the interfaces (ports) connected to it.

```
diagnose npu np6 port-list
Chip  XAUI Ports  Max  Cross-chip
      XAUI      Speed offloading
-----
np6_0  0
      1  port10  1G    Yes
      1  port9   1G    Yes
      1  port12  1G    Yes
      1  port11  1G    Yes
      1  port14  1G    Yes
      1  port13  1G    Yes
      1  port16  1G    Yes
      1  port15  1G    Yes
      1  port5   1G    Yes
      1  port7   1G    Yes
      1  port8   1G    Yes
      1  port6   1G    Yes
      1  port3   1G    Yes
      1  port4   1G    Yes
      1  port1   1G    Yes
      1  port2   1G    Yes
      2
      3
-----
```

FortiGate-500D fast path architecture

The FortiGate-500D includes one NP6 processor connected to eight 1Gb SFP interfaces (port1-port8) and eight 1Gb RJ-45 Ethernet ports (port9-16).

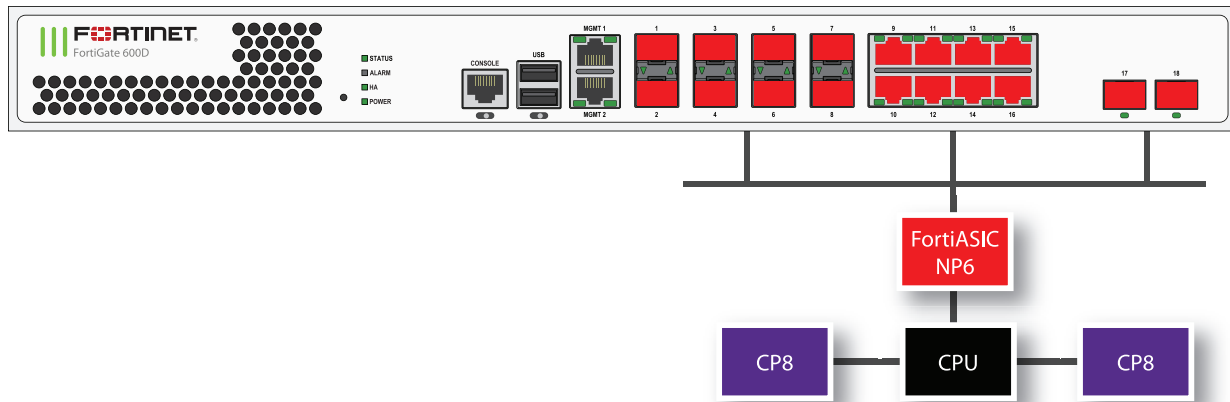


You can use the following command to display the FortiGate-500D NP6 configuration. The command output shows one NP6 named NP6_0 and the interfaces (ports) connected to it.

```
diagnose npu np6 port-list
Chip  XAUI Ports  Max  Cross-chip
      Speed offloading
-----
np6_0  0
      1  port10  1G    Yes
      1  port9   1G    Yes
      1  port12  1G    Yes
      1  port11  1G    Yes
      1  port14  1G    Yes
      1  port13  1G    Yes
      1  port16  1G    Yes
      1  port15  1G    Yes
      1  port5   1G    Yes
      1  port7   1G    Yes
      1  port8   1G    Yes
      1  port6   1G    Yes
      1  port3   1G    Yes
      1  port4   1G    Yes
      1  port1   1G    Yes
      1  port2   1G    Yes
      2
      3
-----
```

FortiGate-600D fast path architecture

The FortiGate-600D includes one NP6 processor connected to eight 1Gb SFP interfaces (port1-port8) and eight 1Gb RJ-45 Ethernet ports (port9-16) and two 10Gb SFP+ interfaces (port17 and port18).



You can use the following command to display the FortiGate-600D NP6 configuration. The command output shows one NP6 named NP6_0 and the interfaces (ports) connected to it.

```
diagnose npu np6 port-list
Chip  XAUI Ports  Max  Cross-chip
      Speed offloading
-----
np6_0  0
      1  port10  1G    Yes
      1  port9   1G    Yes
      1  port12  1G    Yes
      1  port11  1G    Yes
      1  port14  1G    Yes
      1  port13  1G    Yes
      1  port16  1G    Yes
      1  port15  1G    Yes
      1  port5   1G    Yes
      1  port7   1G    Yes
      1  port8   1G    Yes
      1  port6   1G    Yes
      1  port3   1G    Yes
      1  port4   1G    Yes
      1  port1   1G    Yes
      1  port2   1G    Yes
      2  port17  10G   Yes
      3  port18  10G   Yes
-----
```

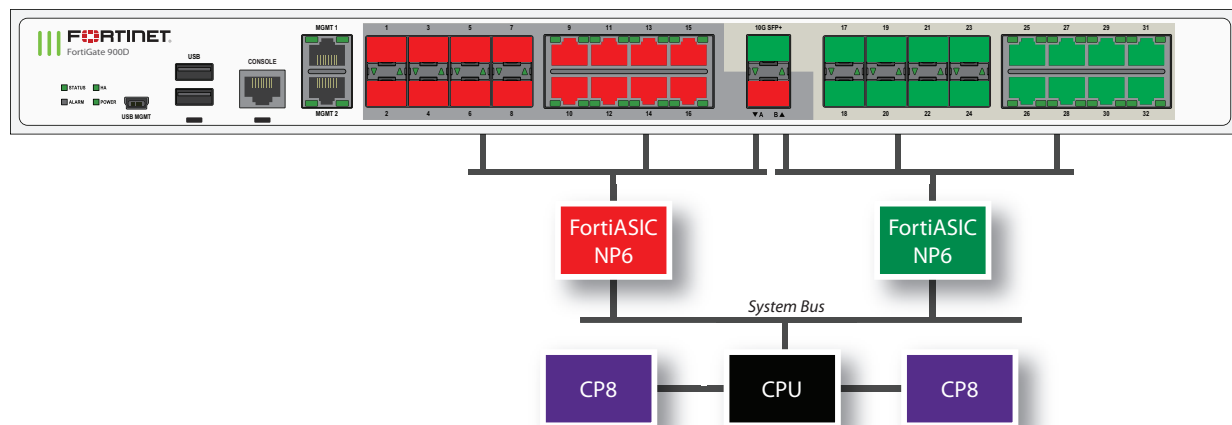
FortiGate-900D fast path architecture

The FortiGate-900D includes two NP6 processors that are not connected by an integrated switch fabric (ISF). Without an ISF, traffic through a FortiGate-900D could experience lower latency than traffic through similar hardware with an ISF. The NP6 processors are connected to network interfaces as follows:



Because the FortiGate-900D does not have an ISF you cannot create Link Aggregation Groups (LAGs) that include interfaces connected to both NP6 processors.

- Eight 1Gb SFP interfaces (port17-port24), eight 1Gb RJ-45 Ethernet interfaces (port25-32) and one 10Gb SFP+ interface (portB) share connections to the first NP6 processor.
- Eight 1Gb SFP interfaces (port1-port8), eight RJ-45 Ethernet interfaces (port9-16) and one 10Gb SFP+ interface (portA) share connections to the second NP6 processor.



You can use the following command to display the FortiGate-900D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1. The output also shows the interfaces (ports) connected to each NP6.

```

diagnose npu np6 port-list
Chip   XAUI Ports   Max   Cross-chip
        Speed offloading
-----
np6_0  0
      1   port17  1G     Yes
      1   port18  1G     Yes
      1   port19  1G     Yes
      1   port20  1G     Yes
      1   port21  1G     Yes
      1   port22  1G     Yes
      1   port23  1G     Yes
      1   port24  1G     Yes
      1   port27  1G     Yes
      1   port28  1G     Yes
      1   port25  1G     Yes
      1   port26  1G     Yes
      1   port31  1G     Yes
      1   port32  1G     Yes
      1   port29  1G     Yes
      1   port30  1G     Yes
      2   portB   10G    Yes
      3
-----
np6_1  0
      1   port1   1G     Yes
      1   port2   1G     Yes
      1   port3   1G     Yes
      1   port4   1G     Yes
      1   port5   1G     Yes
      1   port6   1G     Yes
      1   port7   1G     Yes
      1   port8   1G     Yes
      1   port11  1G     Yes
      1   port12  1G     Yes
      1   port9   1G     Yes
      1   port10  1G     Yes
      1   port15  1G     Yes
      1   port16  1G     Yes
      1   port13  1G     Yes
      1   port14  1G     Yes
      2   portA   10G    Yes
      3

```

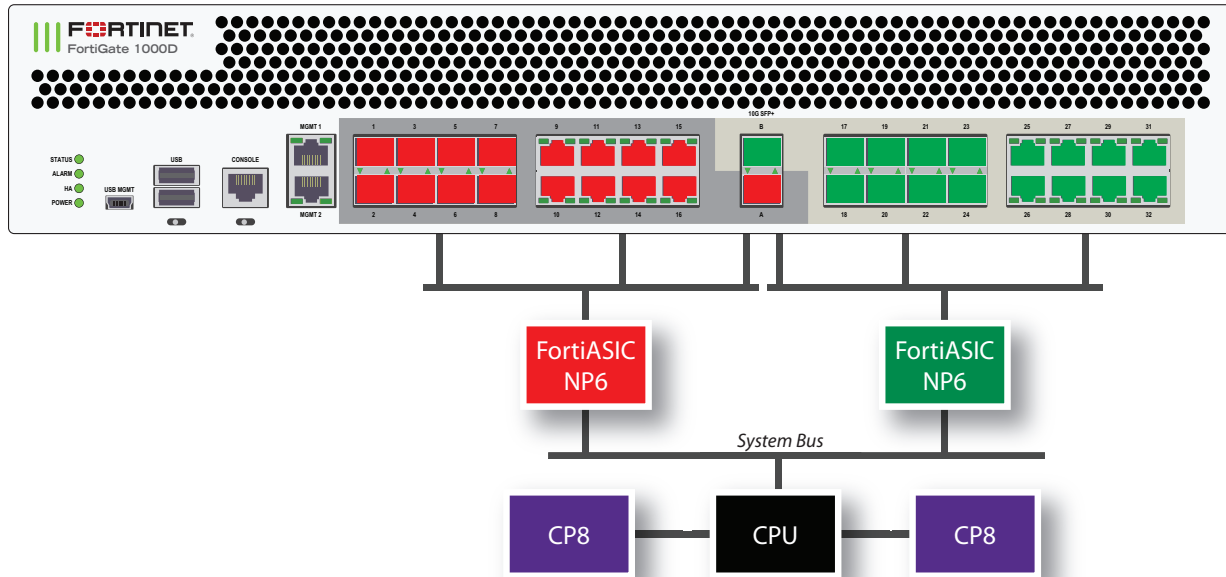
FortiGate-1000D fast path architecture

The FortiGate-1000D includes two NP6 processors that are not connected by an integrated switch fabric (ISF). The NP6 processors are connected to network interfaces as follows:



Because the FortiGate-1000D does not have an ISF you cannot create Link Aggregation Groups (LAGs) that include interfaces connected to both NP6 processors.

- Eight 1Gb SFP interfaces (port17-port24), eight 1Gb RJ-45 Ethernet interfaces (port25-32) and one 10Gb SFP+ interface (portB) share connections to the first NP6 processor.
- Eight 1Gb SFP interfaces (port1-port8), eight RJ-45 Ethernet interfaces (port9-16) and one 10Gb SFP+ interface (portA) share connections to the second NP6 processor.



You can use the following command to display the FortiGate-1000D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1. The output also shows the interfaces (ports) connected to each NP6.

```

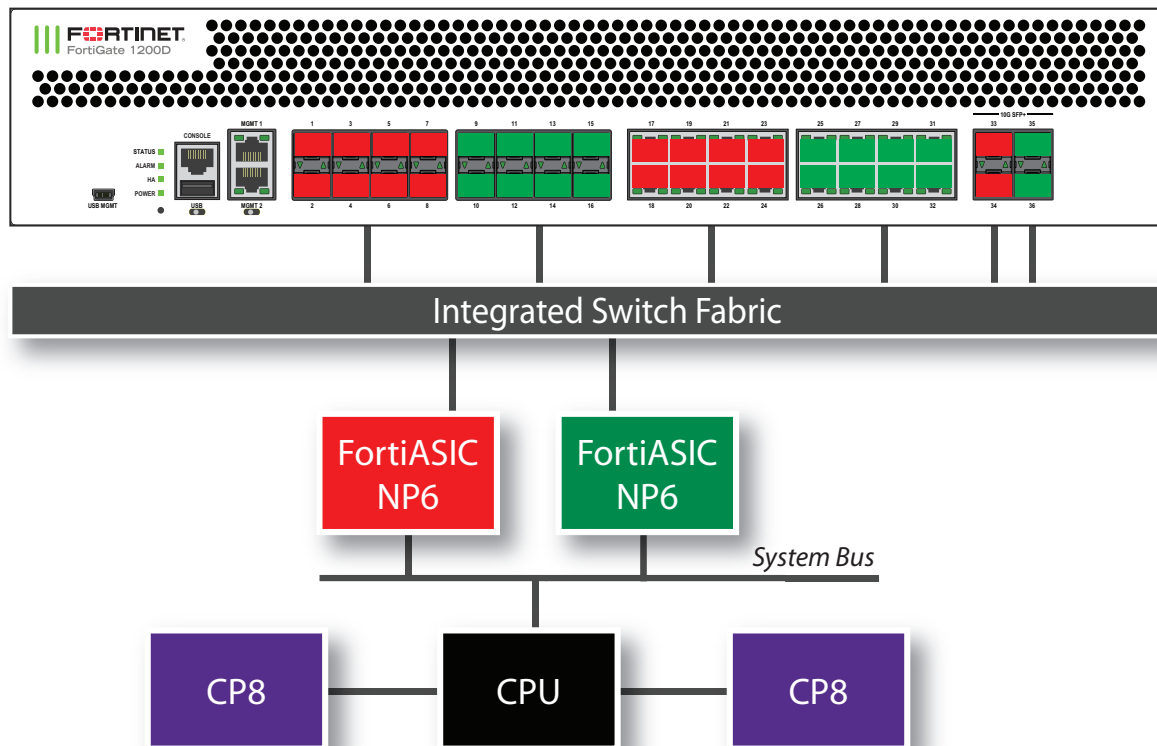
diagnose npu np6 port-list
Chip   XAUI Ports   Max   Cross-chip
        Speed offloading
-----
np6_0  0
      1   port17  1G     Yes
      1   port18  1G     Yes
      1   port19  1G     Yes
      1   port20  1G     Yes
      1   port21  1G     Yes
      1   port22  1G     Yes
      1   port23  1G     Yes
      1   port24  1G     Yes
      1   port27  1G     Yes
      1   port28  1G     Yes
      1   port25  1G     Yes
      1   port26  1G     Yes
      1   port31  1G     Yes
      1   port32  1G     Yes
      1   port29  1G     Yes
      1   port30  1G     Yes
      2   portB   10G    Yes
      3
-----
np6_1  0
      1   port1   1G     Yes
      1   port2   1G     Yes
      1   port3   1G     Yes
      1   port4   1G     Yes
      1   port5   1G     Yes
      1   port6   1G     Yes
      1   port7   1G     Yes
      1   port8   1G     Yes
      1   port11  1G     Yes
      1   port12  1G     Yes
      1   port9   1G     Yes
      1   port10  1G     Yes
      1   port15  1G     Yes
      1   port16  1G     Yes
      1   port13  1G     Yes
      1   port14  1G     Yes
      2   portA   10G    Yes
      3

```

FortiGate-1200D fast path architecture

The FortiGate-1200D features two NP6 processors both connected to an integrated switch fabric.

- Eight SFP 1Gb interfaces (port1-port8), eight RJ-45 Ethernet ports (port17-24) and two SFP+ 10Gb interfaces (port33 and port34) share connections to the first NP6 processor.
- Eight SFP 1Gb interfaces (port9-port16), eight RJ-45 Ethernet ports (port25-32) and two SFP+ 10Gb interfaces (port35-port36) share connections to the second NP6 processor.



You can use the following command to display the FortiGate-1200D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1. The output also shows the interfaces (ports) connected to each NP6.

```

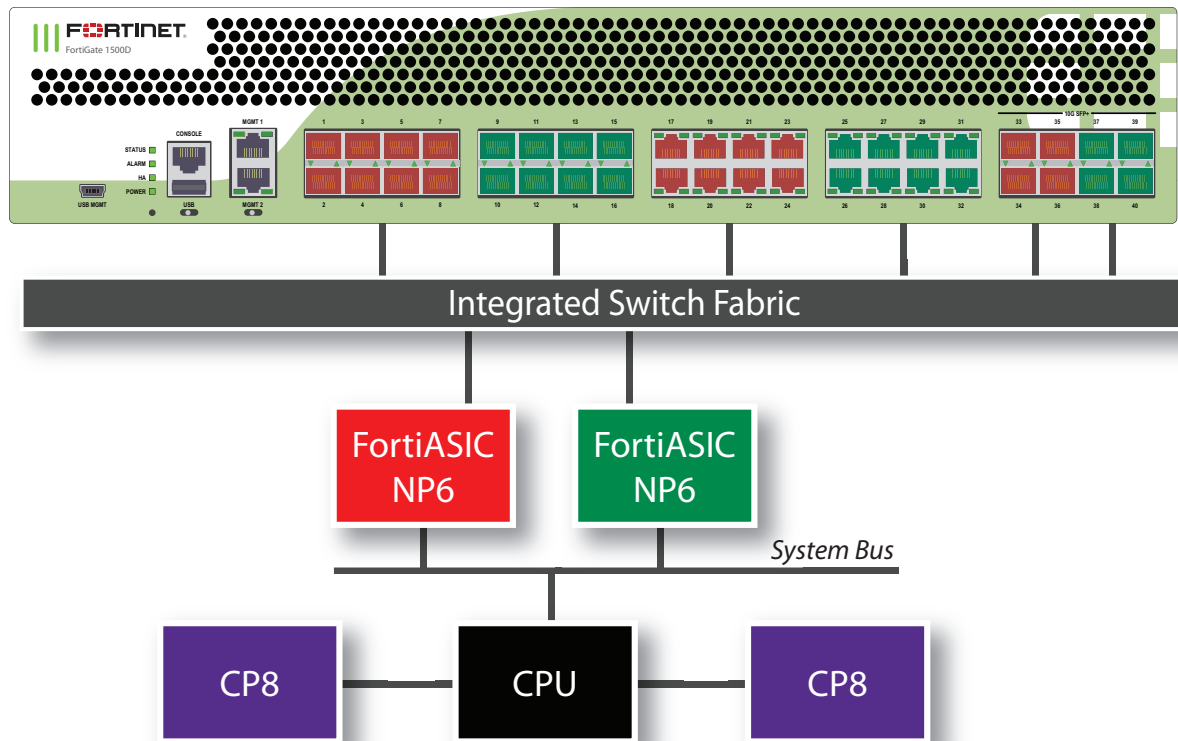
diagnose npu np6 port-list
Chip    XAUI  Ports    Max    Cross-chip
        Speed offloading
-----
np6_0   0      port33    10G    Yes
        1      port34    10G    Yes
        2      port1     1G     Yes
        2      port3     1G     Yes
        2      port5     1G     Yes
        2      port7     1G     Yes
        2      port17    1G     Yes
        2      port19    1G     Yes
        2      port21    1G     Yes
        2      port23    1G     Yes
        3      port2     1G     Yes
        3      port4     1G     Yes
        3      port6     1G     Yes
        3      port8     1G     Yes
        3      port18    1G     Yes
        3      port20    1G     Yes
        3      port22    1G     Yes
        3      port24    1G     Yes
-----
np6_1   0      port35    10G    Yes
        1      port36    10G    Yes
        2      port9     1G     Yes
        2      port11    1G     Yes
        2      port13    1G     Yes
        2      port15    1G     Yes
        2      port25    1G     Yes
        2      port27    1G     Yes
        2      port29    1G     Yes
        2      port31    1G     Yes
        3      port10    1G     Yes
        3      port12    1G     Yes
        3      port14    1G     Yes
        3      port16    1G     Yes
        3      port26    1G     Yes
        3      port28    1G     Yes
        3      port30    1G     Yes
        3      port32    1G     Yes
-----

```

FortiGate-1500D fast path architecture

The FortiGate-1500D features two NP6 processors both connected to an integrated switch fabric.

- Eight SFP 1Gb interfaces (port1-port8), eight RJ-45 Ethernet ports (port17-24) and four SFP+ 10Gb interfaces (port33-port36) share connections to the first NP6 processor.
- Eight SFP 1Gb interfaces (port9-port16), eight RJ-45 Ethernet ports (port25-32) and four SFP+ 10Gb interfaces (port37-port40) share connections to the second NP6 processor.



You can use the following command to display the FortiGate-1500D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1. The output also shows the interfaces (ports) connected to each NP6.

```
diagnose npu np6 port-list
```

Chip	XAUI	Ports	Max Speed	Cross-chip offloading
np6_0	0	port1	1G	Yes
	0	port5	1G	Yes
	0	port17	1G	Yes
	0	port21	1G	Yes
	0	port33	10G	Yes
	1	port2	1G	Yes
	1	port6	1G	Yes
	1	port18	1G	Yes
	1	port22	1G	Yes
	1	port34	10G	Yes
	2	port3	1G	Yes

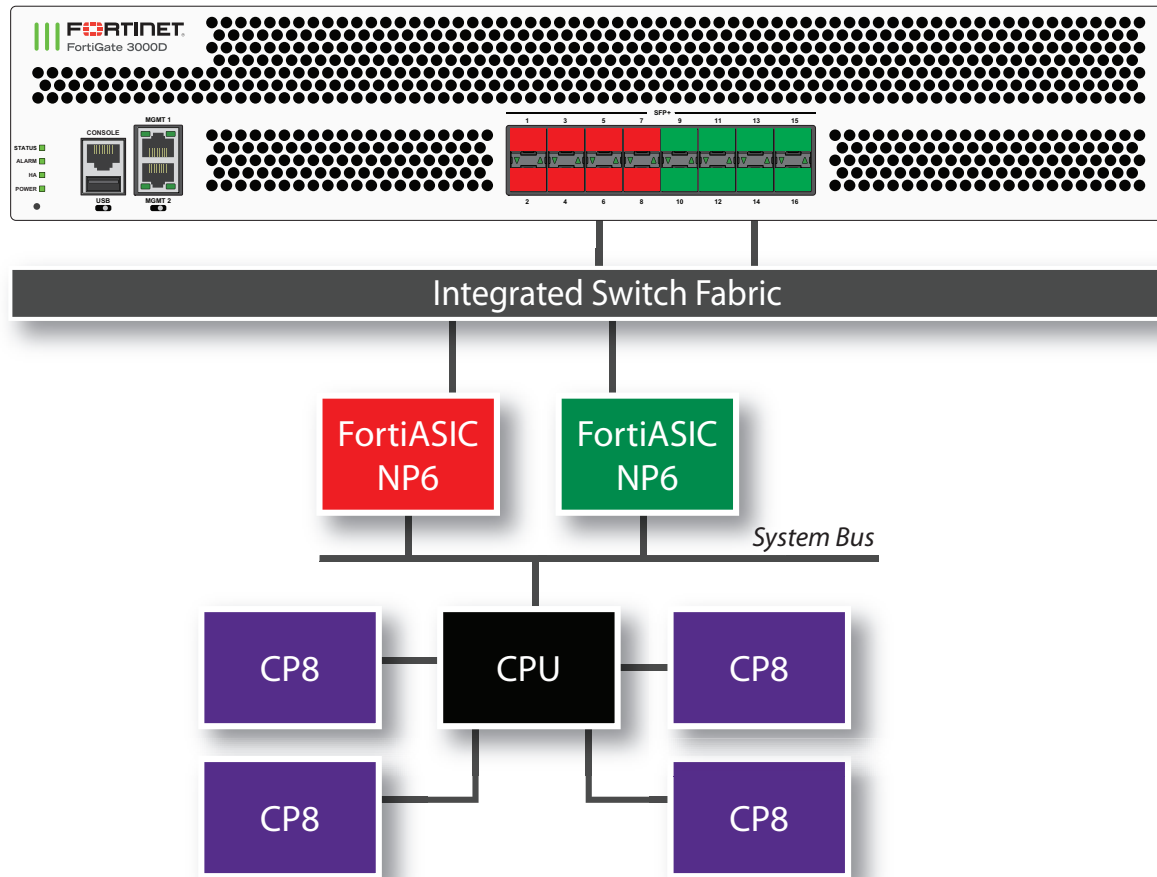
	2	port7	1G	Yes
	2	port19	1G	Yes
	2	port23	1G	Yes
	2	port35	10G	Yes
	3	port4	1G	Yes
	3	port8	1G	Yes
	3	port20	1G	Yes
	3	port24	1G	Yes
	3	port36	10G	Yes

np6_1	0	port9	1G	Yes
	0	port13	1G	Yes
	0	port25	1G	Yes
	0	port29	1G	Yes
	0	port37	10G	Yes
	1	port10	1G	Yes
	1	port14	1G	Yes
	1	port26	1G	Yes
	1	port30	1G	Yes
	1	port38	10G	Yes
	2	port11	1G	Yes
	2	port15	1G	Yes
	2	port27	1G	Yes
	2	port31	1G	Yes
	2	port39	10G	Yes
	3	port12	1G	Yes
	3	port16	1G	Yes
	3	port28	1G	Yes
	3	port32	1G	Yes
	3	port40	10G	Yes

FortiGate-3000D fast path architecture

The FortiGate-3000D features 16 front panel SFP+ 10Gb interfaces connected to two NP6 processors through an Integrated Switch Fabric (ISF). The FortiGate-3000D has the following fastpath architecture:

- 8 SFP+ 10Gb interfaces, port1 through port8 share connections to the first NP6 processor (np6_0).
- 8 SFP+ 10Gb interfaces, port9 through port16 share connections to the second NP6 processor (np6_1).



You can use the following command to display the FortiGate-3000D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1 and the interfaces (ports) connected to each NP6.

```
dia npu np6 port-list
```

Chip	XAUI	Ports	Max Speed	Cross-chip offloading
np6_0	0	port1	10G	Yes
	0	port6	10G	Yes
	1	port2	10G	Yes
	1	port5	10G	Yes
	2	port3	10G	Yes
	2	port8	10G	Yes
	3	port4	10G	Yes
	3	port7	10G	Yes
np6_1	0	port10	10G	Yes
	0	port13	10G	Yes
	1	port9	10G	Yes
	1	port14	10G	Yes
	2	port12	10G	Yes
	2	port15	10G	Yes
	3	port11	10G	Yes

```

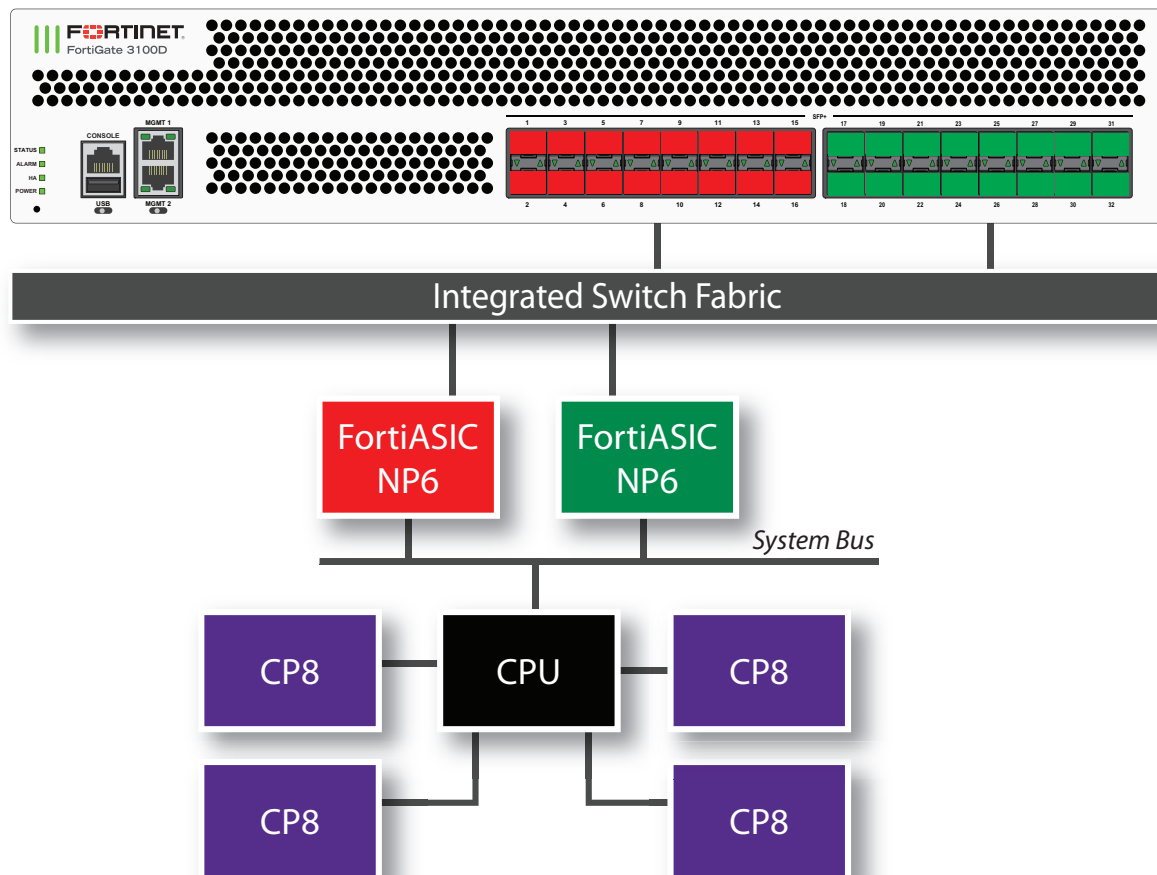
3      port16  10G   Yes
-----

```

FortiGate-3100D fast path architecture

The FortiGate-3100D features 32 SFP+ 10Gb interfaces connected to two NP6 processors through an Integrated Switch Fabric (ISF). The FortiGate-3100D has the following fastpath architecture:

- 16 SFP+ 10Gb interfaces, port1 through port16 share connections to the first NP6 processor (np6_0).
- 16 SFP+ 10Gb interfaces, port27 through port32 share connections to the second NP6 processor (np6_1).



You can use the following command to display the FortiGate-3100D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1 and the interfaces (ports) connected to each NP6.

```

diag npu np6 port-list
Chip  XAUI Ports  Max  Cross-chip
      XAUI Ports  Speed offloading
-----
np6_0  0    port1   10G   Yes
      0    port6   10G   Yes
      0    port10  10G   Yes
      0    port13  10G   Yes

```

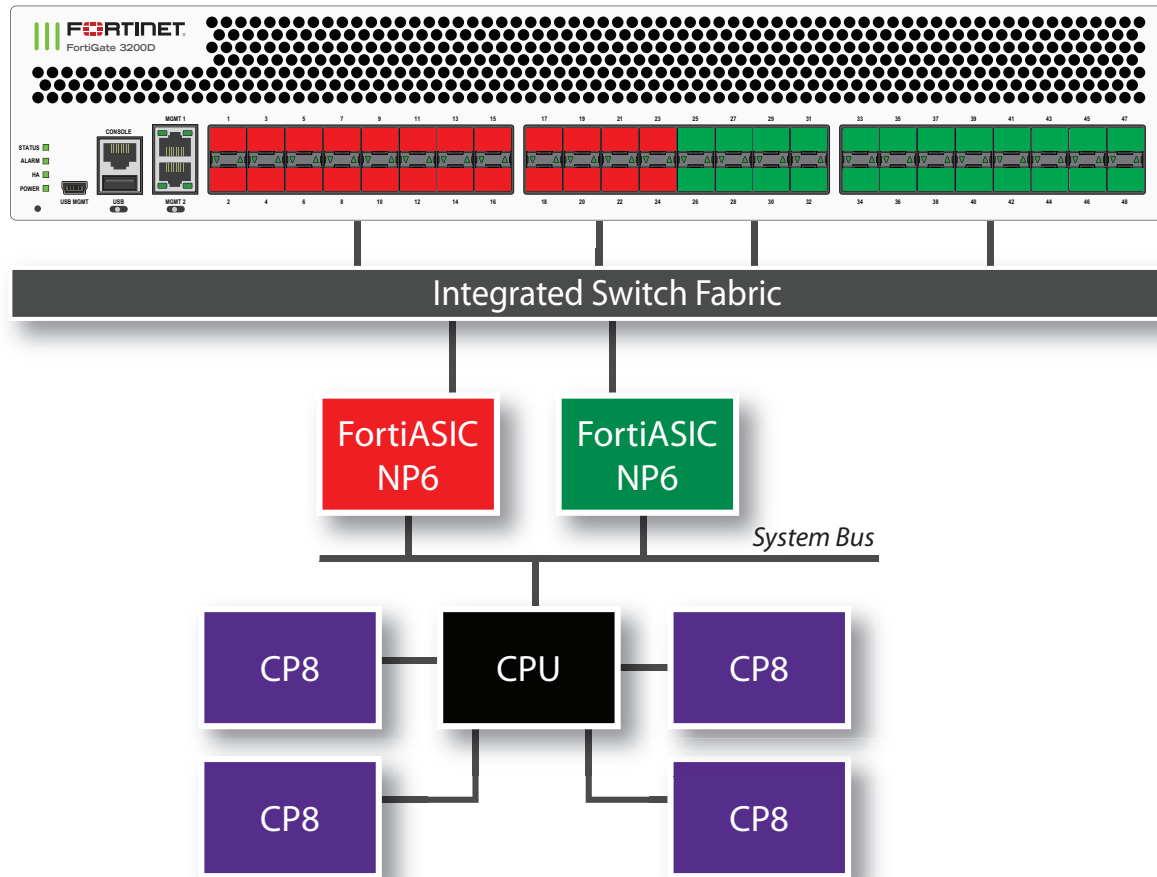
	1	port2	10G	Yes
	1	port5	10G	Yes
	1	port9	10G	Yes
	1	port14	10G	Yes
	2	port3	10G	Yes
	2	port8	10G	Yes
	2	port12	10G	Yes
	2	port15	10G	Yes
	3	port4	10G	Yes
	3	port7	10G	Yes
	3	port11	10G	Yes
	3	port16	10G	Yes

np6_1	0	port17	10G	Yes
	0	port21	10G	Yes
	0	port25	10G	Yes
	0	port29	10G	Yes
	1	port18	10G	Yes
	1	port22	10G	Yes
	1	port26	10G	Yes
	1	port30	10G	Yes
	2	port19	10G	Yes
	2	port23	10G	Yes
	2	port27	10G	Yes
	2	port31	10G	Yes
	3	port20	10G	Yes
	3	port24	10G	Yes
	3	port28	10G	Yes
	3	port32	10G	Yes

FortiGate-3200D fast path architecture

The FortiGate-3200D features two NP6 processors connected to an Integrated Switch Fabric (ISF). The FortiGate-3200D has the following fastpath architecture:

- 24 SFP+ 10Gb interfaces, port1 through port24 share connections to the first NP6 processor (np6_0).
- 24 SFP+ 10Gb interfaces, port25 through port48 share connections to the second NP6 processor (np6_1).



You can use the following command to display the FortiGate-3200D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1 and the interfaces (ports) connected to each NP6.

```
diagnose npu np6 port-list
```

Chip	XAUI	Ports	Max Speed	Cross-chip offloading
np6_0	0	port1	10G	Yes
	0	port5	10G	Yes
	0	port10	10G	Yes
	0	port13	10G	Yes
	0	port17	10G	Yes
	0	port22	10G	Yes
	1	port2	10G	Yes
	1	port6	10G	Yes
	1	port9	10G	Yes
	1	port14	10G	Yes
	1	port18	10G	Yes
	1	port21	10G	Yes
	2	port3	10G	Yes
	2	port7	10G	Yes
	2	port12	10G	Yes
	2	port15	10G	Yes

	2	port19	10G	Yes
	2	port24	10G	Yes
	3	port4	10G	Yes
	3	port8	10G	Yes
	3	port11	10G	Yes
	3	port16	10G	Yes
	3	port20	10G	Yes
	3	port23	10G	Yes

np6_1	0	port26	10G	Yes
	0	port29	10G	Yes
	0	port33	10G	Yes
	0	port37	10G	Yes
	0	port41	10G	Yes
	0	port45	10G	Yes
	1	port25	10G	Yes
	1	port30	10G	Yes
	1	port34	10G	Yes
	1	port38	10G	Yes
	1	port42	10G	Yes
	1	port46	10G	Yes
	2	port28	10G	Yes
	2	port31	10G	Yes
	2	port35	10G	Yes
	2	port39	10G	Yes
	2	port43	10G	Yes
	2	port47	10G	Yes
	3	port27	10G	Yes
	3	port32	10G	Yes
	3	port36	10G	Yes
	3	port40	10G	Yes
	3	port44	10G	Yes
	3	port48	10G	Yes

FortiGate-3700D fast path architecture

The FortiGate-3700D features four NP6 processors. The first two NP6 processors (np6_0 and np6_1) can be configured for low latency operation. The low latency configuration changes the FortiGate-3700D fast path architecture.

FortiGate-3700D low latency fast path architecture

Ports 25 to 32 can be used for low latency offloading. As long as traffic enters and exits the FortiGate-3700D through ports connected to the same NP6 processor and using these low latency ports the traffic will be offloaded and have lower latency than other NP6 offloaded traffic. Latency is reduced by bypassing the integrated switch fabric (ISF).

You can use the following command to turn on low latency mode for np6_0 and np6_1:

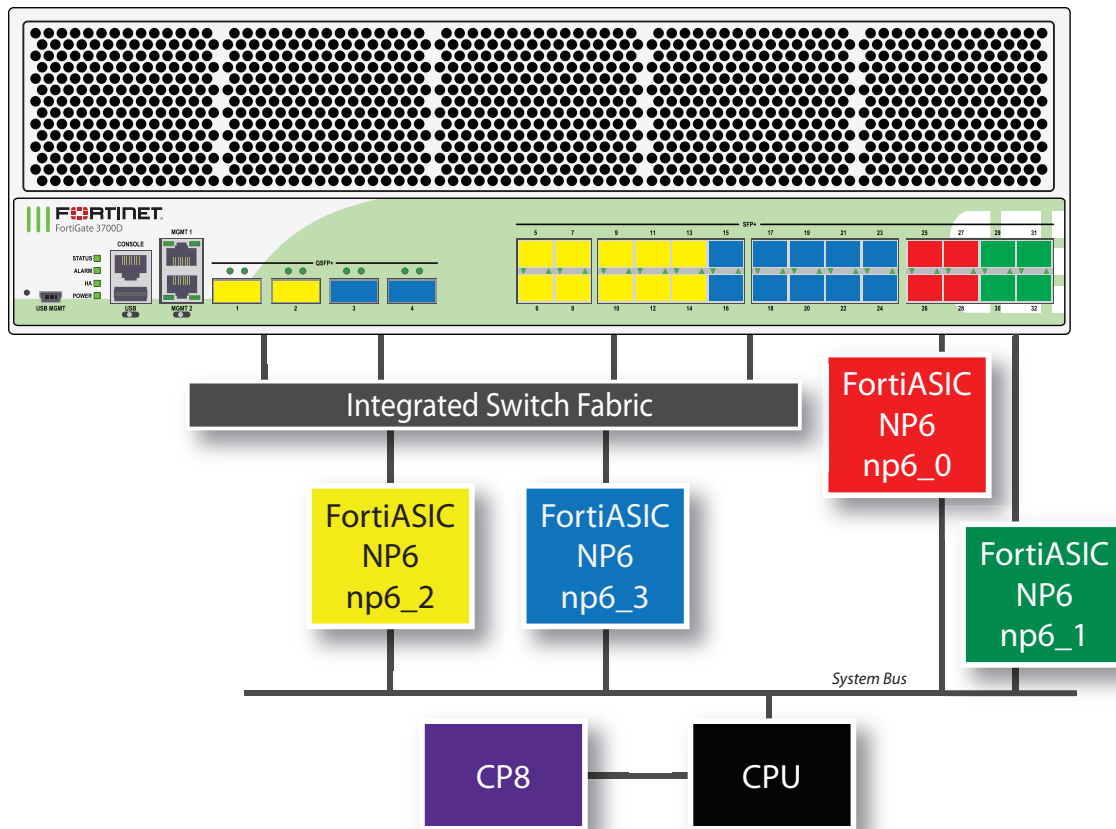
```
config system np6
  edit np6_0
    set low-latency-mode enable
  next
  edit np6_1
    set low-latency-mode enable
end
```



You do not have to turn on low latency to both np6_0 and np6_1. If you turn on low latency for just one NP6, the other NP6 will still be mapped according to the normal latency configuration.

With low latency enabled for both np6_0 and np6_1 the FortiGate-3700D has the following fastpath architecture:

- Four SFP+ 10Gb interfaces, port25 to port28, share connections to the first NP6 processor (np6_0) so sessions entering one of these ports and exiting through another will experience low latency
- Four SFP+ 10Gb interfaces, port29 to port32, share connections to the second NP6 processor (np6_1) so sessions entering one of these ports and exiting through another will experience low latency
- Ten SFP+ 10Gb interfaces, port5 to port14, and two 40Gb QSFP interfaces, port1 and port2, share connections to the third NP6 processor (np6_2).
- Ten SFP+ 10Gb interfaces, port15 to port24, and two 40Gb QSFP interfaces, port3 and port4, share connections to the fourth NP6 processor (np6_3).



You can use the following command to display the FortiGate-3700D NP6 configuration. In this output example, the first two NP6s (np6_0 and np6_1) are configured for low latency. The command output shows four NP6s named NP6_0, NP6_1, NP6_2, and NP6_3 and the interfaces (ports) connected to each NP6.

```
diagnose npu np6 port-list
Chip  XAUI Ports  Max  Cross-chip
      Speed  offloading
-----
np6_2  0    port5    10G  Yes
      0    port9    10G  Yes
      0    port13   10G  Yes
      1    port6    10G  Yes
      1    port10   10G  Yes
      1    port14   10G  Yes
      2    port7    10G  Yes
      2    port11   10G  Yes
      3    port8    10G  Yes
      3    port12   10G  Yes
      0-3  port1    40G  Yes
      0-3  port2    40G  Yes
-----
np6_3  0    port15   10G  Yes
      0    port19   10G  Yes
      0    port23   10G  Yes
      1    port16   10G  Yes
```

	1	port20	10G	Yes
	1	port24	10G	Yes
	2	port17	10G	Yes
	2	port21	10G	Yes
	3	port18	10G	Yes
	3	port22	10G	Yes
	0-3	port3	40G	Yes
	0-3	port4	40G	Yes

np6_0	0	port26	10G	No
	1	port25	10G	No
	2	port28	10G	No
	3	port27	10G	No

np6_1	0	port30	10G	No
	1	port29	10G	No
	2	port32	10G	No
	3	port31	10G	No

FortiGate-3700D normal latency fast path architecture

You can use the following command to turn off low latency mode for np6_0 and np6_1:

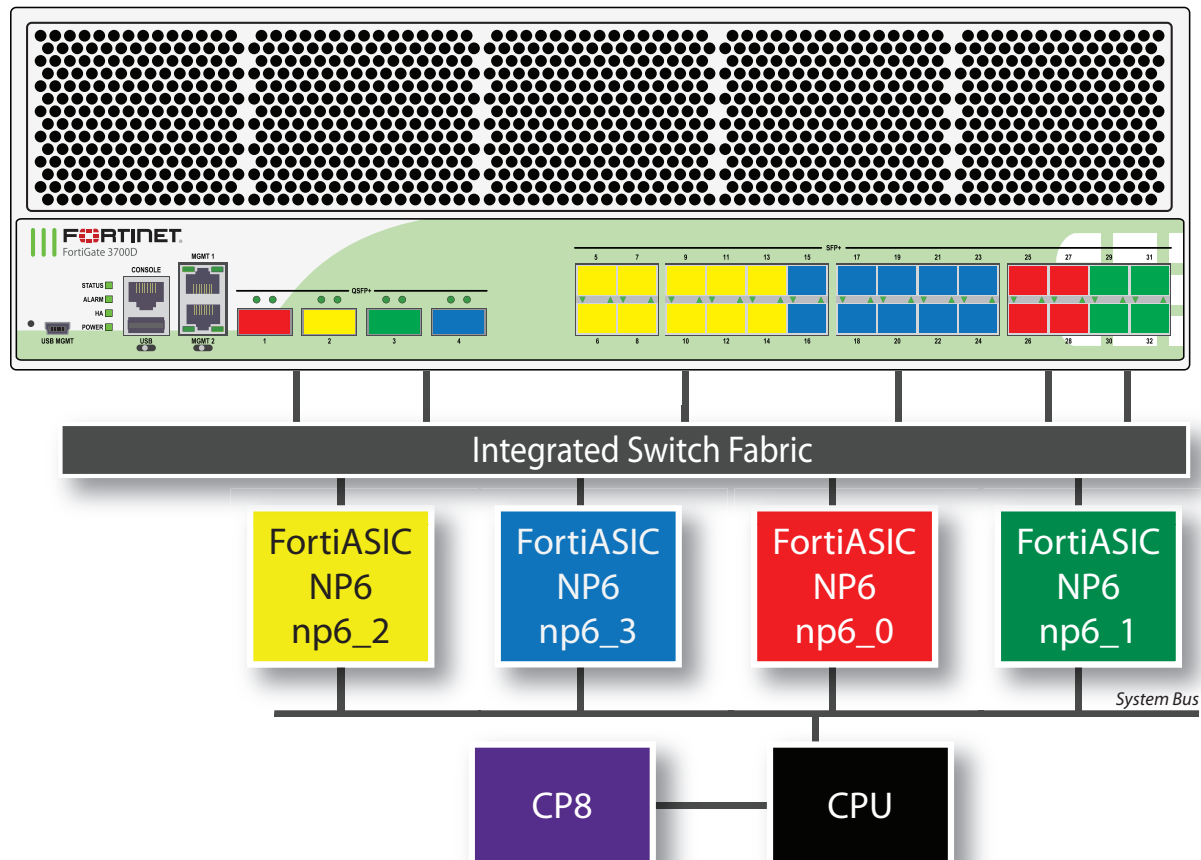
```
config system np6
  edit np6_0
    set low-latency-mode disable
  next
  edit np6_1
    set low-latency-mode disable
end
```



You do not have to turn off low latency to both np6_0 and np6_1. If you turn off low latency to just one NP6, the other NP6 will still be mapped according to the normal configuration.

In addition to turning off low latency, entering these commands also changes how ports are mapped to NP6s. Port1 is now mapped to np6_0 and port 3 is not mapped to np6_1. The FortiGate-3700D has the following fastpath architecture:

- One 40Gb QSFP interface, port1, and four SFP+ 10Gb interfaces, port25 to port28 share connections to the first NP6 processor (np6_0).
- One 40Gb QSFP interface, port3, and four SFP+ 10Gb interfaces, port29 to port32 share connections to the second NP6 processor (np6_1).
- One 40Gb QSFP interface, port2 and ten SFP+ 10Gb interfaces, port5 to port14 share connections to the third NP6 processor (np6_2).
- One 40Gb QSFP interface, port4, and ten SFP+ 10Gb interfaces, port15 to port24 share connections to the fourth NP6 processor (np6_3).



You can use the following command to display the FortiGate-3700D NP6 configuration with low latency turned off for np6_0 and np6_1. The command output shows four NP6s named NP6_0, NP6_1, NP6_2, and NP6_3 and the interfaces (ports) connected to each NP6.

```

dia npu np6 port-list
Chip  XAUI  Ports                Max    Cross-chip
      XAUI  Ports                Speed  offloading
-----
np6_0  0    port26                10G    Yes
      1    port25                10G    Yes
      2    port28                10G    Yes
      3    port27                10G    Yes
      0-3  port1                40G    Yes
-----
np6_1  0    port30                10G    Yes
      1    port29                10G    Yes
      2    port32                10G    Yes
      3    port31                10G    Yes
      0-3  port3                40G    Yes
-----
np6_2  0    port5                 10G    Yes
      0    port9                 10G    Yes
      0    port13                10G    Yes
      1    port6                 10G    Yes

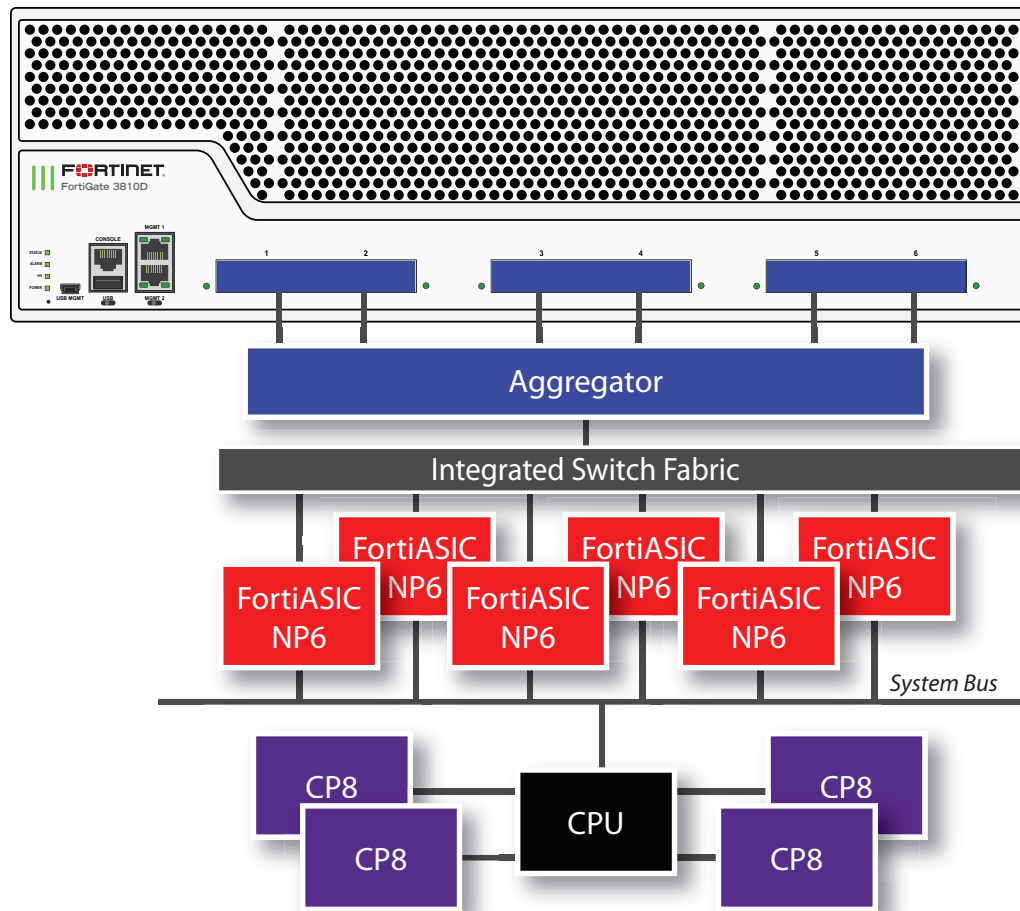
```

	1	port10	10G	Yes
	1	port14	10G	Yes
	2	port7	10G	Yes
	2	port11	10G	Yes
	3	port8	10G	Yes
	3	port12	10G	Yes
	0-3	port2	40G	Yes

np6_3	0	port15	10G	Yes
	0	port19	10G	Yes
	0	port23	10G	Yes
	1	port16	10G	Yes
	1	port20	10G	Yes
	1	port24	10G	Yes
	2	port17	10G	Yes
	2	port21	10G	Yes
	3	port18	10G	Yes
	3	port22	10G	Yes
	0-3	port4	40G	Yes

FortiGate-3810D fast path architecture

The FortiGate-3800D features 6 front panel 100Gigabit interfaces and eight NP6 processors connected to an Integrated Switch Fabric (ISF). Individual interfaces are not mapped to NP6 processors. Instead an Aggregator connects the front panel interfaces to the ISF. Because of the aggregator, no special mapping is required for fastpath offloading.



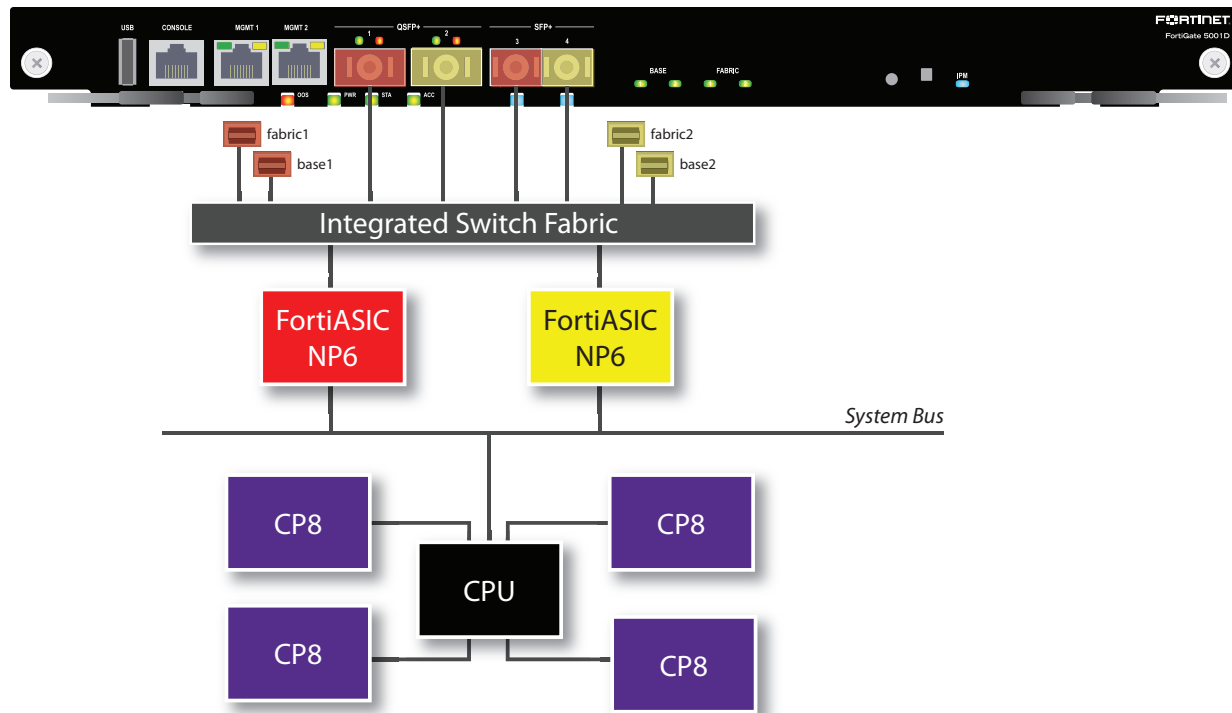
You can use the following command to display the FortiGate-3810D NP6 configuration. The command output shows all NP6s connected to each interface (port) with cross-chip offloading supported for each port.

```
diagnose npu np6 port-list
Chip   XAUI Ports   Max      Cross-chip
      Speed    offloading
-----
all    0-3  port1    100000M Yes
all    0-3  port2    100000M Yes
all    0-3  port3    100000M Yes
all    0-3  port4    100000M Yes
all    0-3  port5    100000M Yes
all    0-3  port6    100000M Yes
-----
```

FortiGate-5001D fast path architecture

The FortiGate5001D features two NP6 processors.

- port1, port3, fabric1 and base1 share connections to the first NP6 processor.
- port2, port4, fabric2 and base2 share connections to the second NP6 processor.



NP6 default interface mapping

You can use the following command to display the FortiGate-5001D NP6 configuration. The command output shows two NP6s named NP6_0 and NP6_1. The output also shows the interfaces (ports) connected to each NP6.

```
diagnose npu np6 port-list
```

Chip	XAUI	Ports	Max Speed	Cross-chip offloading
np6_0	0	port3	10G	Yes
	1			
	2	base1	1G	Yes
	3			
	0-3	port1	40G	Yes
	0-3	fabric1	40G	Yes
	0-3	fabric3	40G	Yes
	0-3	fabric5	40G	Yes
np6_1	0			
	1	port4	10G	Yes
	2			
	3	base2	1G	Yes
	0-3	port2	40G	Yes
	0-3	fabric2	40G	Yes
	0-3	fabric4	40G	Yes

NP6 interface mapping with split ports

If you use the following CLI command to split port1:

```
config system global
    set split-port port1
end
```

The new split ports (port1/1 to port 1/4) are mapped to the same NP6 as the un-split port1 interface:

```
diagnose npu np6 port-list
```

Chip	XAUI	Ports	Max Speed	Cross-chip offloading
np6_0	0	port3	10G	Yes
	0	port1/1	10G	Yes
	1	port1/2	10G	Yes
	2	base1	1G	Yes
	2	port1/3	10G	Yes
	3	port1/4	10G	Yes
	0-3	fabric1	40G	Yes
	0-3	fabric3	40G	Yes
	0-3	fabric5	40G	Yes
np6_1	0			
	1	port4	10G	Yes
	2			
	3	base2	1G	Yes
	0-3	port2	40G	Yes
	0-3	fabric2	40G	Yes
	0-3	fabric4	40G	Yes

NP4 Acceleration

NP4 network processors provide fastpath acceleration by offloading communication sessions from the FortiGate CPU. When the first packet of a new session is received by an interface connected to an NP4 processor, just like any session connecting with any FortiGate interface, the session is forwarded to the FortiGate CPU where it is matched with a security policy. If the session is accepted by a security policy and if the session can be offloaded its session key is copied to the NP4 processor that received the packet. All of the rest of the packets in the session are intercepted by the NP4 processor and fast-pathed out of the FortiGate unit to their destination without ever passing through the FortiGate CPU. The result is enhanced network performance provided by the NP4 processor plus the network processing load is removed from the CPU. In addition, the NP4 processor can handle some CPU intensive tasks, like IPsec VPN encryption/decryption.

Session keys (and IPsec SA keys) are stored in the memory of the NP4 processor that is connected to the interface that received the packet that started the session. All sessions are fast-pathed and accelerated, even if they exit the FortiGate unit through an interface connected to another NP4. The key to making this possible is the Integrated Switch Fabric (ISF) that connects the NP4s and the FortiGate unit interfaces together. The ISF allows any port connectivity. All ports and NP4s can communicate with each other over the ISF.

There are no special ingress and egress fast path requirements because traffic enters and exits on interfaces connected to the same ISF. Most FortiGate models with multiple NP4 processors connect all interfaces and NP4 processors to the same ISF (except management interfaces) so this should not ever be a problem.

There is one limitation to keep in mind; the capacity of each NP4 processor. An individual NP4 processor has a capacity of 20 Gbps (10 Gbps ingress and 10 Gbps egress). Once an NP4 processor hits its limit, sessions that are over the limit are sent to the CPU. You can avoid this problem by as much as possible distributing incoming sessions evenly among the NP4 processors. To be able to do this you need to be aware of which interfaces connect to which NP4 processors and distribute incoming traffic accordingly.

Some FortiGate units contain one NP4 processor with all interfaces connected to it and to the ISF. As a result, offloading is supported for traffic between any pair of interfaces.

Some FortiGate units include NP4Lite processors. These network processors have the same functionality and limitations as NP4 processors but with about half the performance. NP4lite processors can be found in mid-range FortiGate models such as the FortiGate-200D and 240D.

Viewing your FortiGate's NP4 configuration

To list the NP4 network processors on your FortiGate unit, use the following CLI command.

```
get hardware npu np4 list
```

The output lists the interfaces that have NP4 processors. For example, for a FortiGate-5001C:

```
get hardware npu np4 list
ID      Model      Slot      Interface
0       On-board    port1 port2 port3 port4
        fabric1 base1 npu0-vlink0 npu0-vlink1
1       On-board    port5 port6 port7 port8
        fabric2 base2 npu1-vlink0 npu1-vlink1
```

NP4lite CLI commands (disabling NP4Lite offloading)

If your FortiGate unit includes an NP4Lite processor the following commands will be available:

- Use the following command to disable or enable NP4Lite offloading. By default NP4lite offloading is enabled. If you want to disable NP4Lite offloading to diagnose a problem enter:

```
diagnose npu nplite fastpath disable
```

This command disables NP4Lite offloading until your FortiGate reboots. You can also re-enable offloading by entering the following command:

```
diagnose npu nplite fastpath enable
```

- NP4lite debug command. Use the following command to debug NP4Lite operation:

```
diagnose npl npl_debug {<parameters>}
```

Configuring NP4 traffic offloading

Offloading traffic to a network processor requires that the FortiGate unit configuration and the traffic itself is suited to hardware acceleration. There are requirements for path the sessions and the individual packets.

NP4 session fast path requirements

Sessions must be fast path ready. Fast path ready session characteristics are:

- Layer 2 type/length must be 0x0800 (IEEE 802.1q VLAN specification is supported)
- Layer 3 protocol must be IPv4
- Layer 4 protocol must be UDP, TCP or ICMP
- Layer 3 / Layer 4 header or content modification must not require a session helper (for example, SNAT, DNAT, and TTL reduction are supported, but application layer content modification is not supported)
- Firewall policies must not include proxy-based or flow-based security features (antivirus, web filtering, email filtering, IPS, application control, or DLP)
- Origin must not be local host (the FortiGate unit)



If you disable anomaly checks by Intrusion Prevention (IPS), you can still enable NP4 hardware accelerated anomaly checks using the `fp-anomaly` field of the `config system interface` CLI command. See [Offloading NP4 anomaly detection on page 70](#)

If a session is not fast path ready, the FortiGate unit will not send the session key to the network processor(s). Without the session key, all session key lookup by a network processor for incoming packets of that session fails, causing all session packets to be sent to the FortiGate unit's main processing resources, and processed at normal speeds.

If a session is fast path ready, the FortiGate unit will send the session key to the network processor(s). Session key lookup then succeeds for subsequent packets from the known session.

Packet fast path requirements

Packets within the session must then also meet packet requirements.

- Incoming packets must not be fragmented.
- Outgoing packets must not require fragmentation to a size less than 385 bytes. Because of this requirement, the configured MTU (Maximum Transmission Unit) for network processors' network interfaces must also meet or exceed the network processors' supported minimum MTU of 385 bytes.

If packet requirements are not met, an individual packet will use FortiGate unit main processing resources, regardless of whether other packets in the session are offloaded to the specialized network processor(s).

In some cases, due to these requirements, a protocol's session(s) may receive a mixture of offloaded and non-offloaded processing.

For example, FTP uses two connections: a control connection and a data connection. The control connection requires a session helper, and cannot be offloaded, but the data connection does not require a session helper, and can be offloaded. Within the offloadable data session, fragmented packets will not be offloaded, but other packets will be offloaded.

Some traffic types differ from general offloading requirements, but still utilize some of the network processors' encryption and other capabilities. Exceptions include IPsec traffic and active-active high availability (HA) load balanced traffic.

Mixing fast path and non-fast path traffic

If packet requirements are not met, an individual packet will be processed by the FortiGate CPU regardless of whether other packets in the session are offloaded to the NP4.

Also, in some cases, a protocol's session(s) may receive a mixture of offloaded and non-offloaded processing. For example, VoIP control packets may not be offloaded but VoIP data packets (voice packets) may be offloaded.

Increasing NP4 offloading capacity using link aggregation groups (LAGs)

NP4 processors can offload sessions received by interfaces in link aggregation groups (LAGs) (IEEE 802.3ad). A LAG combines more than one physical interface into a group that functions like a single interface with a higher capacity than a single physical interface. For example, you could use a LAG if you want to offload sessions on a 3 Gbps link by adding three 1Gbps interfaces to the same LAG.

All offloaded traffic types are supported by LAGs, including IPsec VPN traffic. Just like with normal interfaces, traffic accepted by a LAG is offloaded by the NP4 processor connected to the interfaces in the LAG that receive the traffic to be offloaded. If all interfaces in a LAG are connected to the same NP4 processor, traffic received by that LAG is offloaded by that NP4 processor. The amount of traffic that can be offloaded is limited by the capacity of the NP4 processor.

If a FortiGate has two or more NP4 processors connected by an integrated switch fabric (ISF), you can use LAGs to increase offloading by sharing the traffic load across multiple NP4 processors. You do this by adding physical interfaces connected to different NP4 processors to the same LAG.

Adding a second NP4 processor to a LAG effectively doubles the offloading capacity of the LAG. Adding a third further increases offloading. The actual increase in offloading capacity may not actually be doubled by adding a

second NP4 or tripled by adding a third. Traffic and load conditions and other factors may limit the actual offloading result.

The increase in offloading capacity offered by LAGs and multiple NP4s is supported by the ISF that allows multiple NP4 processors to share session information. On models that have more than one NP4 and no ISF, if you attempt to add interfaces connected to different NP4 processors to a LAG the system displays an error message.

There are also a few limitations to LAG NP4 offloading support for IPsec VPN:

- IPsec VPN anti-replay protection cannot be used if IPsec is configured on a LAG that has interfaces connected to multiple NP4 processors.
- Using a LAG connected to multiple NP4 processors for decrypting incoming IPsec VPN traffic may cause some of the incoming traffic to be decrypted by the CPU. So this configuration is not recommended since not all decryption is offloaded. (Using a LAG connected to multiple NP4 processors for encrypting outgoing IPsec VPN traffic is supported with no limitations.)
- Because the encrypted traffic for one IPsec VPN tunnel has the same 5-tuple, the traffic from one tunnel can only be balanced to one interface in a LAG. This limits the maximum throughput for one IPsec VPN tunnel in an NP4 LAG group to 1Gbps.

NP4 traffic shaping offloading

Accelerated Traffic shaping is supported with the following limitations.

- NP4 processors support policy-based traffic shaping. However, fast path traffic and traffic handled by the FortiGate CPU (slow path) are controlled separately, which means the policy setting on fast path does not consider the traffic on the slow path.
- The port based traffic policing as defined by the inbandwidth and outbandwidth CLI commands is not supported.
- DSCP configurations are supported.
- Per-IP traffic shaping is supported.
- QoS in general is not supported.

You can also use the traffic shaping features of the FortiGate unit's main processing resources by disabling NP4 offloading. See [Disabling NP offloading for firewall policies on page 18](#).

NP4 IPsec VPN offloading

NP4 processors improve IPsec tunnel performance by offloading IPsec encryption and decryption.

Requirements for hardware accelerated IPsec encryption or decryption are a modification of general offloading requirements. Differing characteristics are:

- Origin can be local host (the FortiGate unit)
- In Phase 1 configuration, Local Gateway IP must be specified as an IP address of a network interface for a port attached to a network processor
- SA must have been received by the network processor
- in Phase 2 configuration:
 - encryption algorithm must be DES, 3DES, AES-128, AES-192, AES-256, or null

- authentication must be MD5, SHA1, or null
- if encryption is null, authentication must not also be null
- if replay detection is enabled, `enc-offload-antireplay` must also be `enable` in the CLI



If replay detection is enabled in the Phase 2 configuration, you can enable or disable IPsec encryption and decryption offloading from the CLI. Performance varies by those CLI options and the percentage of packets requiring encryption or decryption. For details, see [NP4 IPsec VPN offloading on page 61](#)

To apply hardware accelerated encryption and decryption, the FortiGate unit's main processing resources must first perform Phase 1 negotiations to establish the security association (SA). The SA includes cryptographic processing instructions required by the network processor, such as which encryption algorithms must be applied to the tunnel. After ISAKMP negotiations, the FortiGate unit's main processing resources send the SA to the network processor, enabling the network processor to apply the negotiated hardware accelerated encryption or decryption to tunnel traffic.

Possible accelerated cryptographic paths are:

- IPsec decryption offload
 - Ingress ESP packet > Offloaded decryption > Decrypted packet egress (fast path)
 - Ingress ESP packet > Offloaded decryption > Decrypted packet to FortiGate unit's main processing resources
- IPsec encryption offload
 - Ingress packet > Offloaded encryption > Encrypted (ESP) packet egress (fast path)
 - Packet from FortiGate unit's main processing resources > Offloaded encryption > Encrypted (ESP) packet egress

NP4 IPsec VPN offloading configuration example

Hardware accelerated IPsec processing, involving either partial or full offloading, can be achieved in either tunnel or interface mode IPsec configurations.

To achieve offloading for both encryption and decryption:

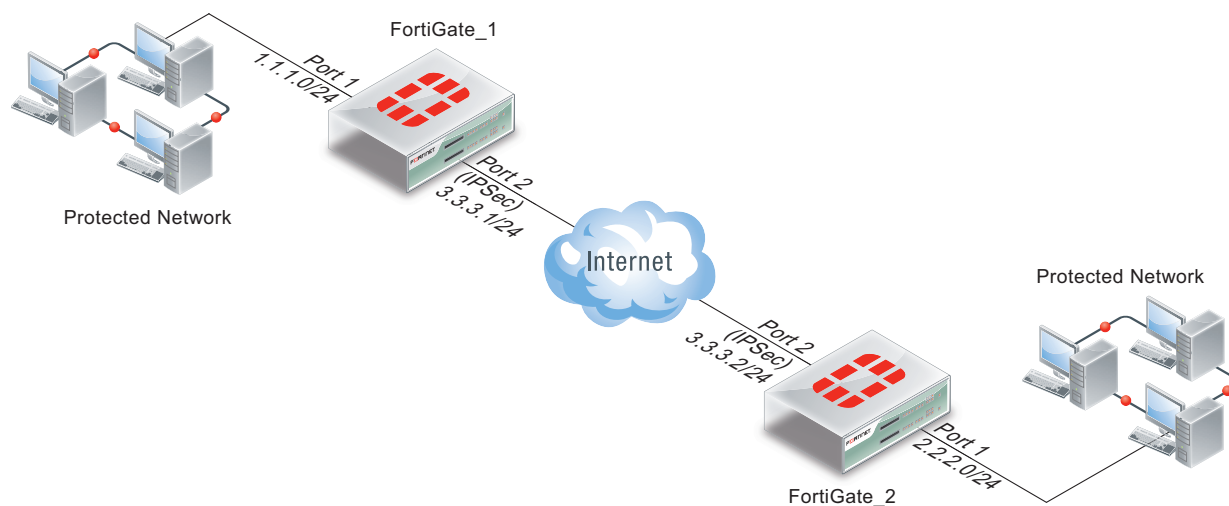
- In Phase 1 configuration's Advanced section, Local Gateway IP must be specified as an IP address of a network interface associated with a port attached to a network processor. (In other words, if Phase 1's Local Gateway IP is Main Interface IP, or is specified as an IP address that is not associated with a network interface associated with a port attached to a network processor, IPsec network processing is not offloaded.)
- In Phase 2 configuration's P2 Proposal section, if the checkbox "Enable replay detection" is enabled, `enc-offload-antireplay` and `dec-offload-antireplay` must be set to `enable` in the CLI.
- `offload-ipsec-host` must be set to `enable` in the CLI.

This section contains example IPsec configurations whose IPsec encryption and decryption processing is hardware accelerated by an NP4 unit contained in a FortiGate-5001B at both ends of the VPN tunnel.



Hardware accelerated IPsec VPN does not require both tunnel endpoints to have the same network processor model. However, if hardware is not symmetrical, the packet forwarding rate is limited by the slower side.

Example network topology for offloaded IPsec processing



Example ports and IP addresses for offloaded IPsec processing

	FortiGate_1		FortiGate_2	
	Port	IP	Port	IP
IPsec tunnel	FortiGate-5001B port 2	3.3.3.1/24	FortiGate-5001B port 2	3.3.3.2/24
Protected network	FortiGate-5001B port 1	1.1.1.0/24	FortiGate-5001B port 1	2.2.2.0/24

Accelerated policy mode IPsec configuration

The following steps create a hardware accelerated policy mode IPsec tunnel between two FortiGate-5001B units, each containing two NP4 processors, the first of which will be used.

To configure hardware accelerated policy mode IPsec

1. On FortiGate_1, go to **VPN > IPsec > Auto Key (IKE)**.
2. Configure Phase 1.
For tunnel mode IPsec and for hardware acceleration, specifying the Local Gateway IP is required. Select **Advanced**. In the Local Gateway IP section, select **Specify** and type the VPN IP address 3.3.3.2, which is the IP address of FortiGate_2's FortiGate-ASM-FB4 module port 2.
3. Configure Phase 2.
4. Select **Enable replay detection**.
5. Use the following command to enable offloading antireplay packets:

```
config system npu
    set enc-offload-antireplay enable
end
```

For details on encryption and decryption offloading options available in the CLI, see "Configuring NP accelerated VPN encryption/decryption offloading".

6. Go to **Policy > Policy > Policy.**

7. Configure a policy to apply the Phase 1 IPsec tunnel you configured in step 2 to traffic between FortiGate-5001B ports 1 and 2.

8. Go to **Router > Static > Static Route.**

9. Configure a static route to route traffic destined for FortiGate_2's protected network to VPN IP address of FortiGate_2's VPN gateway, 3.3.3.2, through the FortiGate-5001B port2.

You can also configure the static route using the following CLI command:

```
config router static
    edit 2
        set device "AMC-SW1/2"
        set dst 2.2.2.0 255.255.255.0
        set gateway 3.3.3.2
    end
```

10. On FortiGate_2, go to **VPN > IPsec > Auto Key (IKE).**

11. Configure Phase 1.

For tunnel mode IPsec and for hardware acceleration, specifying the Local Gateway IP is required.

Select Advanced. In the Local Gateway IP section, select Specify and type the VPN IP address 3.3.3.1, which is the IP address of FortiGate_1's port2.

12. Configure Phase 2.

13. Select **Enable replay detection.**

14. Use the following command to enable offloading antireplay packets:

```
config system npu
    set enc-offload-antireplay enable
end
```

For details on encryption and decryption offloading options available in the CLI, see [Configuring NP accelerated IPsec VPN encryption/decryption offloading on page 16](#).

15. Go to **Policy > Policy > Policy.**

16. Configure a policy to apply the Phase 1 IPsec tunnel you configured in step 9 to traffic between FortiGate-5001B ports 1 and 2.

17. Go to **Router > Static > Static Route.**

18. Configure a static route to route traffic destined for FortiGate_1's protected network to VPN IP address of FortiGate_1's VPN gateway, 3.3.3.1, through the FortiGate-5001B port2.

You can also configure the static route using the following CLI commands:

```
config router static
    edit 2
        set device "AMC-SW1/2"
        set dst 1.1.1.0 255.255.255.0
        set gateway 3.3.3.1
    end
```

19. Activate the IPsec tunnel by sending traffic between the two protected networks.

To verify tunnel activation, go to **VPN > Monitor > IPsec Monitor**.

Accelerated interface mode IPsec configuration

The following steps create a hardware accelerated interface mode IPsec tunnel between two FortiGate units, each containing a FortiGate-ASM-FB4 module.

To configure hardware accelerated interface mode IPsec

1. On FortiGate_1, go to **VPN > IPsec > Auto Key (IKE)**.

2. Configure Phase 1.

For interface mode IPsec and for hardware acceleration, the following settings are required.

- Select **Advanced**.
- Enable the checkbox "Enable IPsec Interface Mode."
- In the Local Gateway IP section, select **Specify** and type the VPN IP address 3.3.3.2, which is the IP address of FortiGate_2's port 2.

3. Configure Phase 2.

4. Select **Enable replay detection**.

5. Use the following command to enable offloading antireplay packets:

```
config system npu
    set enc-offload-antireplay enable
end
```

For details on encryption and decryption offloading options available in the CLI, see "Configuring NP accelerated VPN encryption/decryption offloading".

6. Go to **Policy > Policy > Policy**.

7. Configure two policies (one for each direction) to apply the Phase 1 IPsec configuration you configured in step 2 to traffic leaving from or arriving on FortiGate-ASM-FB4 module port 1.

8. Go to **Router > Static > Static Route**.

9. Configure a static route to route traffic destined for FortiGate_2's protected network to the Phase 1 IPsec device, FGT_1_IPsec.

You can also configure the static route using the following CLI commands:

```
config router static
    edit 2
        set device "FGT_1_IPsec"
        set dst 2.2.2.0 255.255.255.0
    end
```

10. On FortiGate_2, go to **VPN > IPsec > Auto Key (IKE)**.

11. Configure Phase 1.

For interface mode IPsec and for hardware acceleration, the following settings are required.

- Enable the checkbox "Enable IPsec Interface Mode."
- In the Local Gateway IP section, select **Specify** and type the VPN IP address 3.3.3.1, which is the IP address of FortiGate_1's FortiGate-5001B port 2.

12. Configure Phase 2.

13. Select **Enable replay detection**.

14. Use the following command to enable offloading antireplay packets:

```
config system npu
    set enc-offload-antireplay enable
```

end

For details on encryption and decryption offloading options available in the CLI, see "[Hardware acceleration overview](#)" on page 8.

15. Go to **Policy > Policy > Policy**.
16. Configure two policies (one for each direction) to apply the Phase 1 IPsec configuration you configured in step 9 to traffic leaving from or arriving on FortiGate-5001B port 1.
17. Go to **Router > Static > Static Route**.
18. Configure a static route to route traffic destined for FortiGate_1's protected network to the Phase 1 IPsec device, FGT_2_IPsec.
You can also configure the static route using the following CLI commands:

```
config router static
  edit 2
    set device "FGT_2_IPsec"
    set dst 1.1.1.0 255.255.255.0
  next
end
```

19. Activate the IPsec tunnel by sending traffic between the two protected networks.
To verify tunnel activation, go to **VPN > Monitor > IPsec Monitor**.

Configuring Inter-VDOM link acceleration with NP4 processors

FortiGate units with NP4 processors include inter-VDOM links that can be used to accelerate inter-VDOM link traffic.



Traffic is blocked if you enable IPS for traffic passing over inter-VDOM links if that traffic is being offloaded by an NP4 processor. If you disable NP4 offloading traffic will be allowed to flow. You can disable offloading in individual firewall policies by disabling `auto-asic-offload` for individual policies. You can also use the following command to disable all IPS offloading

```
config ips global
  set np-accel-mode {none | basic}
  set cp-accel-mode {none | basic | advanced}
end
```

- For a FortiGate unit with two NP4 processors there are also two inter-VDOM links, each with two interfaces:
 - **npu0-vlink:**
npu0-vlink0
npu0-vlink1
 - **npu1-vlink:**
npu1-vlink0
npu1-vlink1

These interfaces are visible from the GUI and CLI. For a FortiGate unit with NP4 interfaces, enter the following CLI command (output shown for a FortiGate-5001B):

```
get hardware npu np4 list
ID      Model      Slot      Interface
0       On-board      port1 port2 port3 port4
fabric1 base1 npu0-vlink0 npu0-vlink1
```

```
1      On-board          port5 port6 port7 port8
                        fabric2 base2 npu1-vlink0 npu1-vlink1
```

By default the interfaces in each inter-VDOM link are assigned to the root VDOM. To use these interfaces to accelerate inter-VDOM link traffic, assign each interface in a pair to the VDOMs that you want to offload traffic between. For example, if you have added a VDOM named New-VDOM to a FortiGate unit with NP4 processors, you can go to **System > Network > Interfaces** and edit the **npu0-vlink1** interface and set the **Virtual Domain** to **New-VDOM**.

This results in an inter-VDOM link between root and New-VDOM. You can also do this from the CLI:

```
config system interface
  edit npu0-vlink1
    set vdom New-VDOM
  end
```

Using VLANs to add more accelerated Inter-VDOM links

You can add VLAN interfaces to the accelerated inter-VDOM links to create inter-VDOM links between more VDOMs. For the links to work, the VLAN interfaces must be added to the same inter-VDOM link, must be on the same subnet, and must have the same VLAN ID.

For example, to accelerate inter-VDOM link traffic between VDOMs named Marketing and Engineering using VLANs with VLAN ID 100 go to **System > Network > Interfaces** and select **Create New** to create the VLAN interface associated with the Marketing VDOM:

Name	Marketing-link
Type	VLAN
Interface	npu0-vlink0
VLAN ID	100
Virtual Domain	Marketing
IP/Network Mask	172.20.120.12/24

Create the inter-VDOM link associated with Engineering VDOM:

Name	Engineering-link
Type	VLAN
Interface	npu0-vlink1
VLAN ID	100
Virtual Domain	Engineering
IP/Network Mask	172.20.120.22/24

Or do the same from the CLI:

```

config system interface
  edit Marketing-link
    set vdom Marketing
    set ip 172.20.120.12/24
    set interface npu0-vlink0
    set vlanid 100
  next
  edit Engineering-link
    set vdom Engineering
    set ip 172.20.120.22/24
    set interface npu0-vlink1
    set vlanid 100

```

Confirm that the traffic is accelerated

Use the following CLI commands to obtain the interface index and then correlate them with the session entries. In the following example traffic was flowing between new accelerated inter-VDOM links and physical ports port1 and port 2 also attached to the NP4 processor.

diagnose ip address list

```

IP=172.31.17.76->172.31.17.76/255.255.252.0 index=5 devname=port1
IP=10.74.1.76->10.74.1.76/255.255.252.0 index=6 devname=port2
IP=172.20.120.12->172.20.120.12/255.255.255.0 index=55 devname=IVL-VLAN1_ROOT
IP=172.20.120.22->172.20.120.22/255.255.255.0 index=56 devname=IVL-VLAN1_VDOM1

```

diagnose sys session list

```

session info: proto=1 proto_state=00 duration=282 expire=24 timeout=0 session info:
  proto=1 proto_state=00 duration=124 expire=59 timeout=0 flags=00000000
  sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/
state=may_dirty npu
statistic(bytes/packets/allow_err): org=180/3/1 reply=120/2/1 tuples=2
origin->sink: org pre->post, reply pre->post dev=55->5/5->55
  gwy=172.31.19.254/172.20.120.22
hook=post dir=org act=snat 10.74.2.87:768->10.2.2.2:8(172.31.17.76:62464)
hook=pre dir=reply act=dnat 10.2.2.2:62464->172.31.17.76:0(10.74.2.87:768)
misc=0 policy_id=4 id_policy_id=0 auth_info=0 chk_client_info=0 vd=0
serial=0000004e tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
per_ip_bandwidth meter: addr=10.74.2.87, bps=880
npu_state=00000000
npu info: flag=0x81/0x81, offload=4/4, ips_offload=0/0, epid=160/218, ipid=218/160,
vlan=32769/0

session info: proto=1 proto_state=00 duration=124 expire=20 timeout=0 flags=00000000
  sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/
state=may_dirty npu
statistic(bytes/packets/allow_err): org=180/3/1 reply=120/2/1 tuples=2
origin->sink: org pre->post, reply pre->post dev=6->56/56->6
  gwy=172.20.120.12/10.74.2.87

```

```

hook=pre dir=org act=noop 10.74.2.87:768->10.2.2.2:8(0.0.0.0:0)
hook=post dir=reply act=noop 10.2.2.2:768->10.74.2.87:0(0.0.0.0:0)
misc=0 policy_id=3 id_policy_id=0 auth_info=0 chk_client_info=0 vd=1
serial=0000004d tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
per_ip_bandwidth meter: addr=10.74.2.87, bps=880
npu_state=00000000
npu info: flag=0x81/0x81, offload=4/4, ips_offload=0/0, epid=219/161, ipid=161/219,
          vlan=0/32769
total session 2

```

Offloading NP4 anomaly detection

Network interfaces associated with a port attached to an NP4 processor can be configured to offload anomaly checking to the NP4 processor. This anomaly checking happens before other offloading and separately from DoS policy anomaly checking. Using the following command, each FortiGate interface can have a different anomaly checking configuration even if they are connected to the same NP4 processor.



The options available for this command apply anomaly checking for NP4 sessions in the same way as the command described in [Configuring individual NP6 processors on page 28](#) applies anomaly checking for for NP6 sessions.

```

config system interface
    edit <port-name>
        set fp-anomaly <anomalies>
    end

```

where <anomalies> can be one, more than one or all of the following:

Anomaly	Description
drop_icmp_frag	Drop ICMP fragments to pass.
drop_icmpland	Drop ICMP Land.
drop_ipland	Drop IP Land.
drop_iplsrr	Drop IP with Loose Source Record Route option.
drop_iprr	Drop IP with Record Route option.
drop_ipsecurity	Drop IP with Security option.
drop_ipssrr	Drop IP with Strict Source Record Route option.
drop_ipstream	Drop IP with Stream option.
drop_iptimestamp	Drop IP with Timestamp option.

Anomaly	Description
drop_ipunknown_option	Drop IP with malformed option.
drop_ipunknown_prot	Drop IP with Unknown protocol.
drop_tcp_fin_noack	Drop TCP FIN with no ACK flag set to pass.
drop_tcp_no_flag	Drop TCP with no flag set to pass.
drop_tcp_land	Drop TCP Land.
drop_udp_land	Drop UDP Land.
drop_winnuke	Drop TCP WinNuke.
pass_icmp_frag	Allow ICMP fragments to pass.
pass_icmpland	Allow ICMP Land to pass.
pass_ip_land	Allow IP land to pass.
pass_ip_lsr	Allow IP with Loose Source Record Route option to pass.
pass_ip_rr	Allow IP with Record Route option to pass.
pass_ip_security	Allow IP with Security option to pass.
pass_ip_ssrr	Allow IP with Strict Source Record Route option to pass.
pass_ip_stream	Allow IP with Stream option to pass.
pass_ip_timestamp	Allow IP with Timestamp option to pass.
pass_ipunknown_option	Allow IP with malformed option to pass.
pass_ipunknown_prot	Allow IP with Unknown protocol to pass.
pass_tcp_fin_noack	Allow TCP FIN with no ACK flag set to pass.
pass_tcp_no_flag	Allow TCP with no flag set to pass.
pass_tcp_land	Allow TCP Land to pass.

Anomaly	Description
pass_udpland	Allow UDP Land to pass.
pass_winnuke	Allow TCP WinNuke to pass.

Example

You might configure an NP4 to drop packets with TCP WinNuke or unknown IP protocol anomalies, but to pass packets with an IP time stamp, using hardware acceleration provided by the network processor.

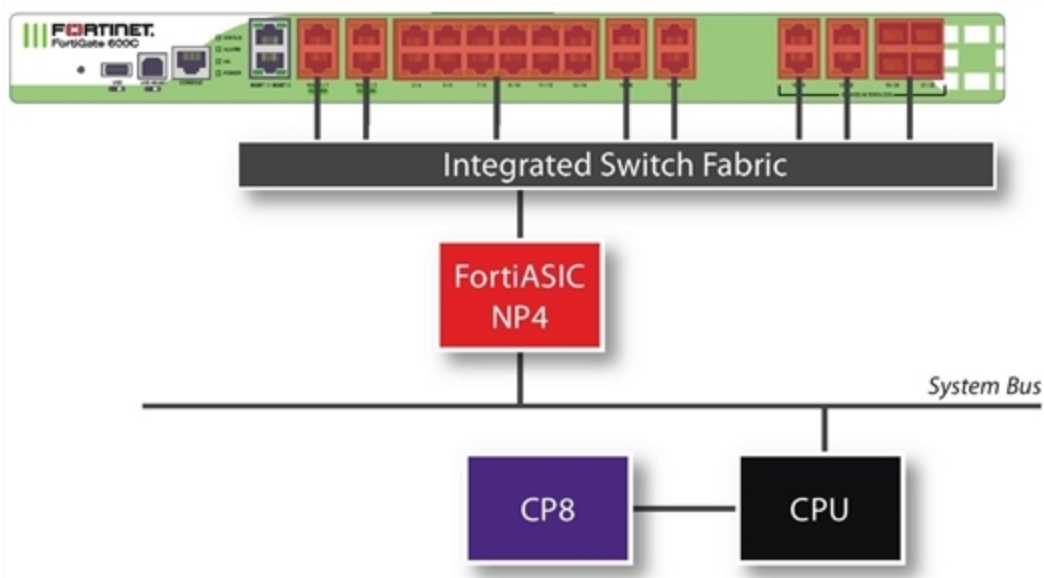
```
config system interface
  edit port1
    set fp-anomaly drop_winnuke drop_ipunknown_prot pass_iptimestamp
  end
```


FortiGate NP4 architectures

This chapter shows the NP4 architecture for the all FortiGate units and modules that include NP4 processors.

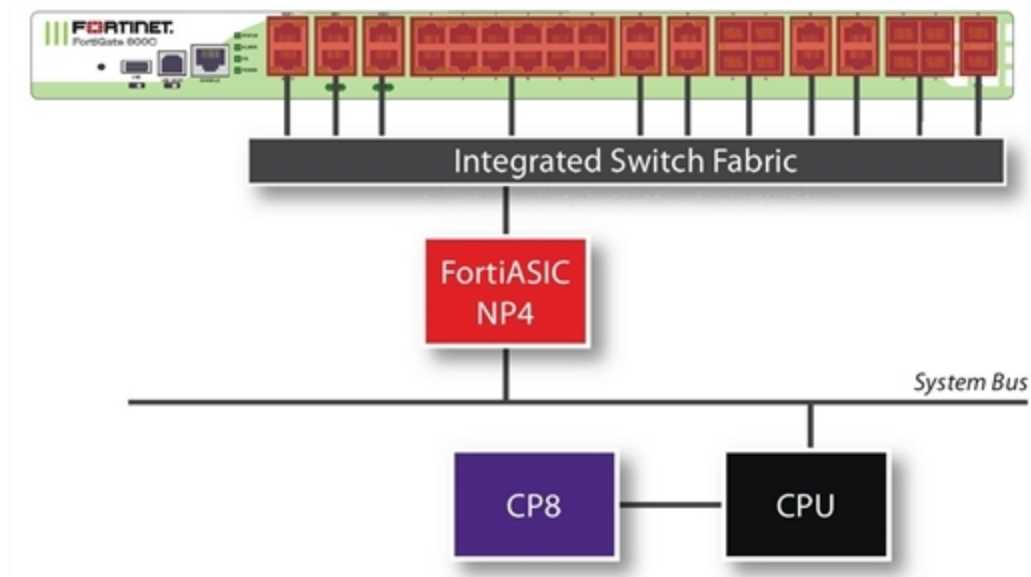
FortiGate-600C

The FortiGate-600C features one NP4 processor. All the ports are connected to this NP4 over the Integrated Switch Fabric. Port1 and port2 are dual failopen redundant RJ-45 ports. Port3-port22 are RJ-45 ethernet ports, and there are four 1Gb SFP interface ports duplicating the port19-port22 connections.



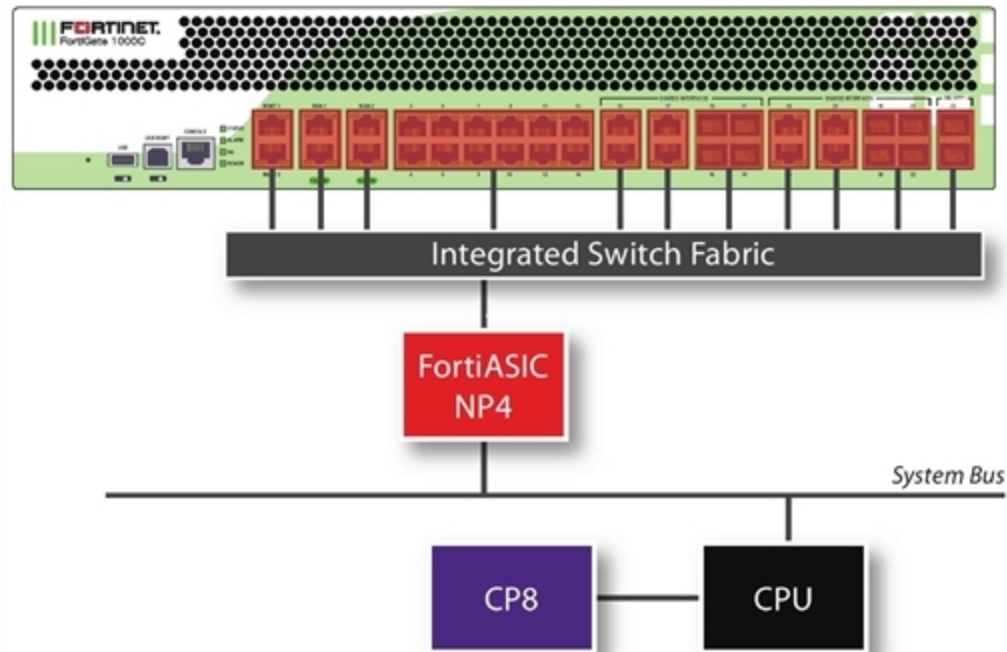
FortiGate-800C

The FortiGate-800C features one NP4 processor. All the ports are connected to this NP4. Port1 and port2 are dual failopen redundant RJ-45 ports. Port3-port22 are RJ-45 ethernet ports, and there are eight 1Gb SFP interface ports duplicating the port15-18 and port19-port22 connections. There are also two 10Gb SFP+ ports, port23 and port24.



FortiGate-1000C

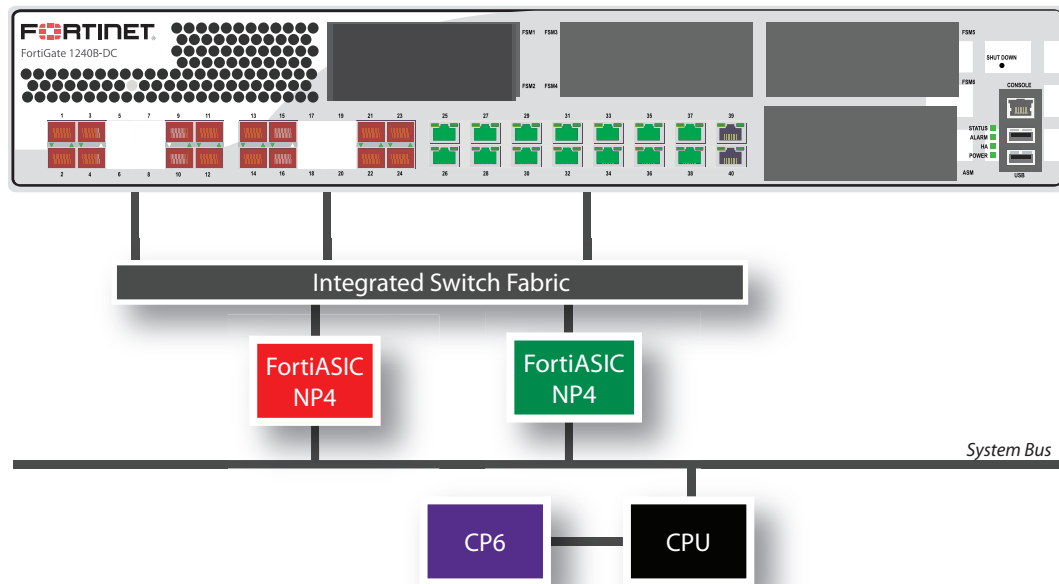
The FortiGate-1000C features one NP4 processor. All the ports are connected to this NP4. Port1 and port2 are dual failopen redundant RJ-45 ports. Port3-port22 are RJ-45 ethernet ports, and there are eight 1Gb SFP interface ports duplicating the port15-18 and port19-port22 connections. There are also two 10Gb SFP+ ports, port23 and port24.



FortiGate-1240B

The FortiGate-1240B features two NP4 processors:

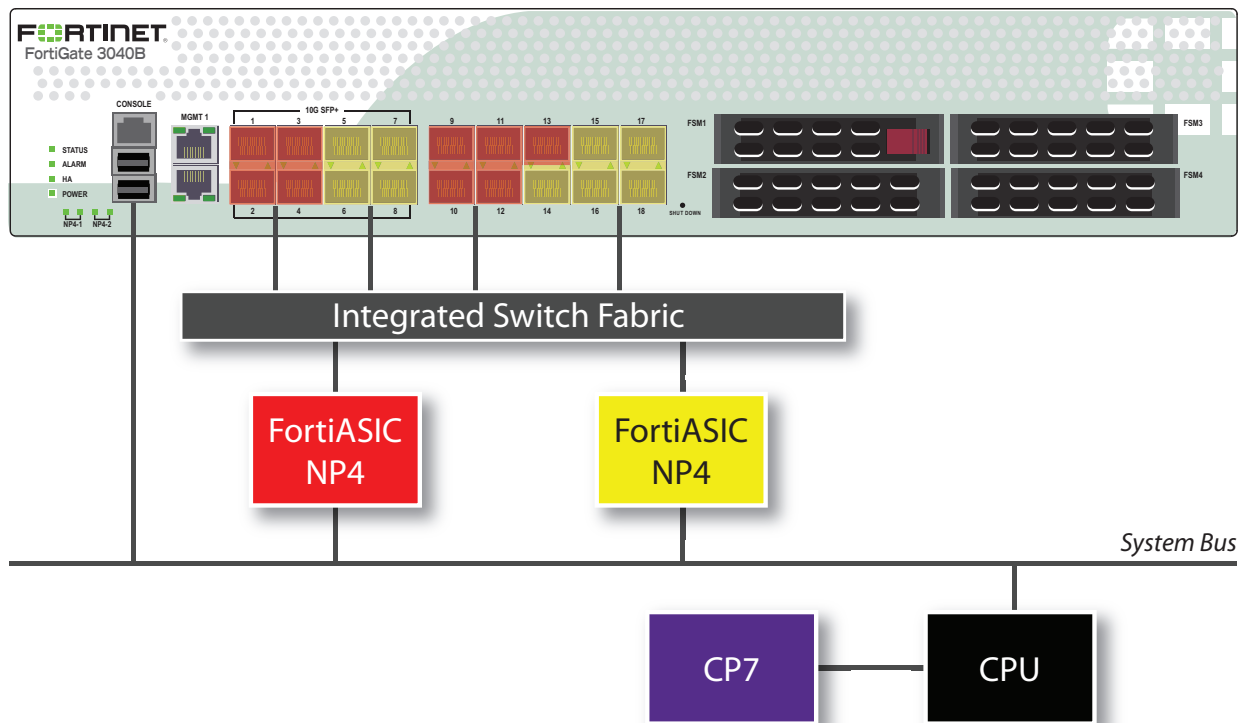
- Port1 to port24 are 1Gb SFP interfaces connected to one NP4 processor.
- Port25 to port38 are RJ-45 ethernet ports, connected to the other NP4 processor.
- Port39 and port40 are not connected to an NP4 processor.



FortiGate-3040B

The FortiGate-3040B features two NP4 processors:

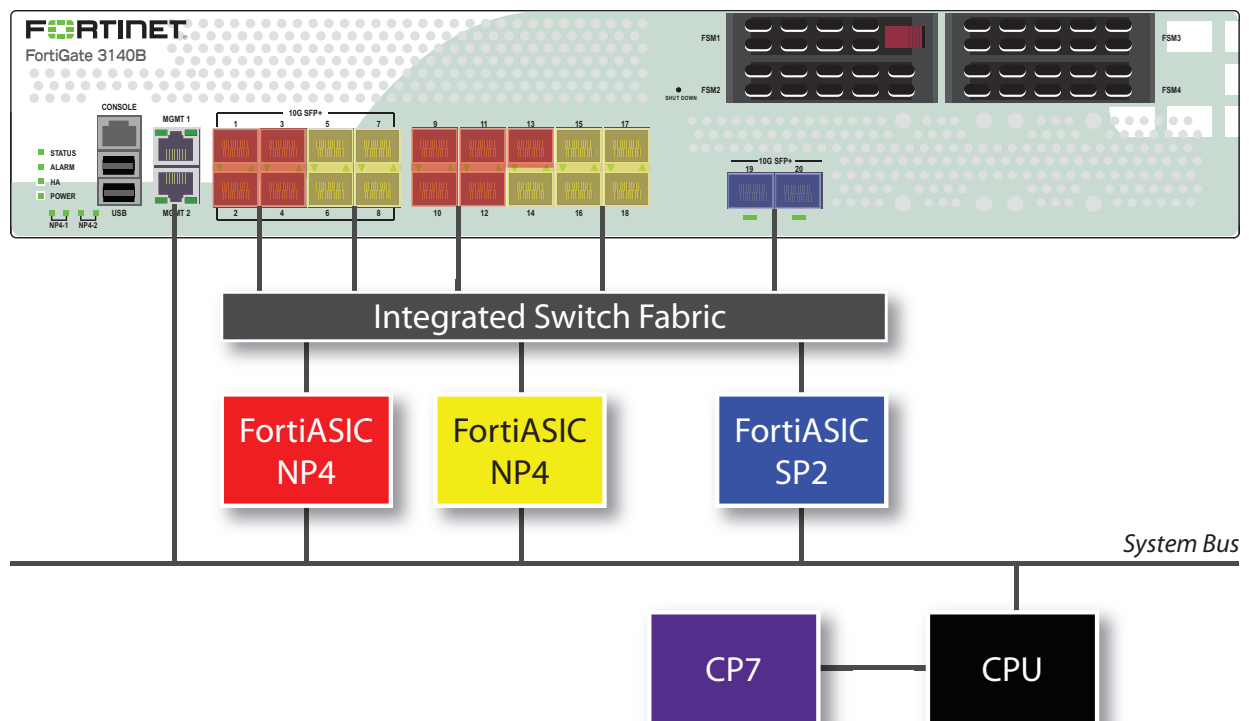
- The 10Gb interfaces, port1, port2, port3, port4, and the 1Gb interfaces, port9, port10, port11, port12, port13, share connections to one NP4 processor.
- The 10Gb interfaces, port5, port6, port7, port8, and the 1Gb interfaces, port14, port15, port16, port17, port18, share connections to the other NP4 processor.



FortiGate-3140B

The FortiGate-3140B features two NP4 processors and one SP2 processor:

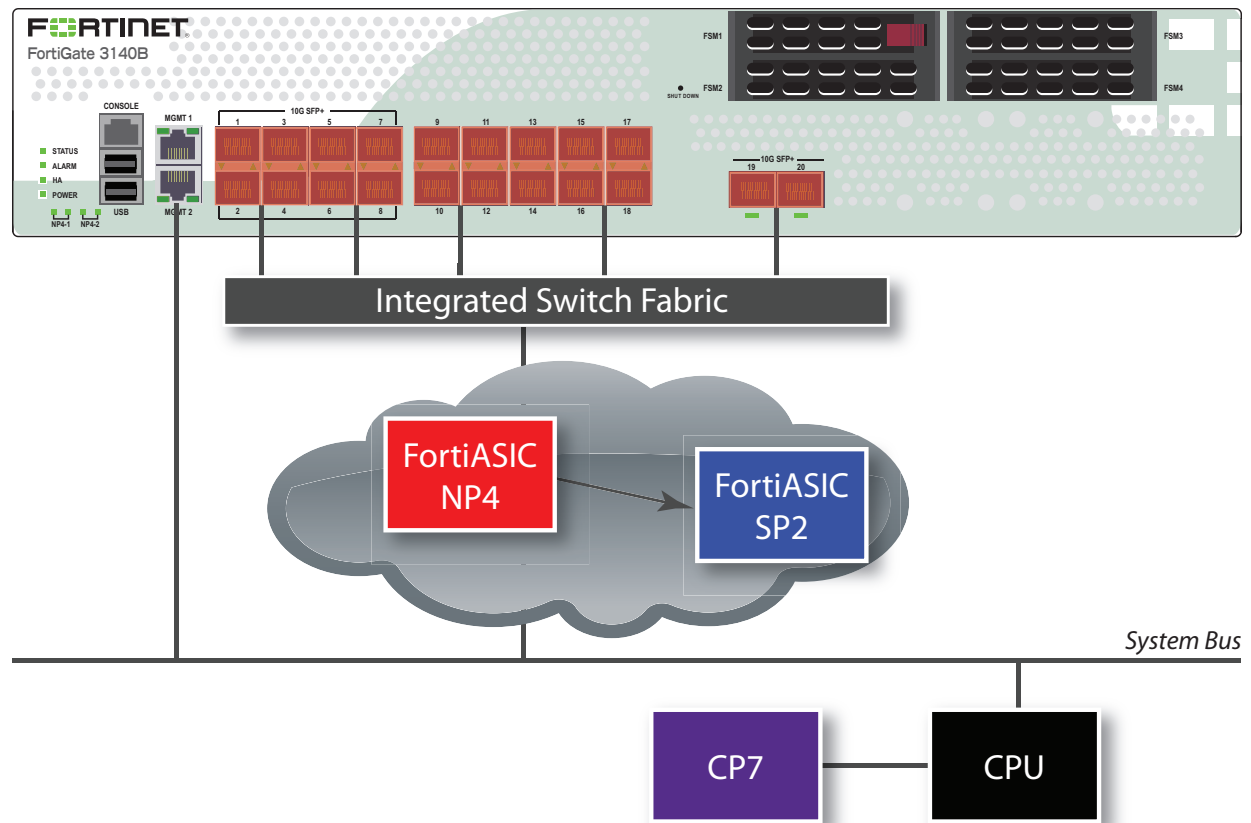
- The 10Gb interfaces, port1, port2, port3, port4, and the 1Gb interfaces, port9, port10, port11, port12, port13, share connections to one NP4 processor.
- The 10Gb interfaces, port5, port6, port7, port8, and the 1Gb interfaces, port14, port15, port16, port17, port18, share connections to the other NP4 processor.
- The 10Gb interfaces, port19 and port20, share connections to the SP2 processor.



FortiGate-3140B — load balance mode

The FortiGate-3140B load balance mode allows you increased flexibility in how you use the interfaces on the FortiGate unit. When enabled, traffic between any two interfaces (excluding management and console) is accelerated. Traffic is not limited to entering and leaving the FortiGate unit in specific interface groupings to benefit from NP4 and SP2 acceleration. You can use any pair of interfaces.

Security acceleration in this mode is limited, however. Only IPS scanning is accelerated in load balance mode.



To enable this feature, issue this CLI command.

```
config system global
    set sp-load-balance enable
end
```

The FortiGate unit will then restart.

To return to the default mode, issue this CLI command.

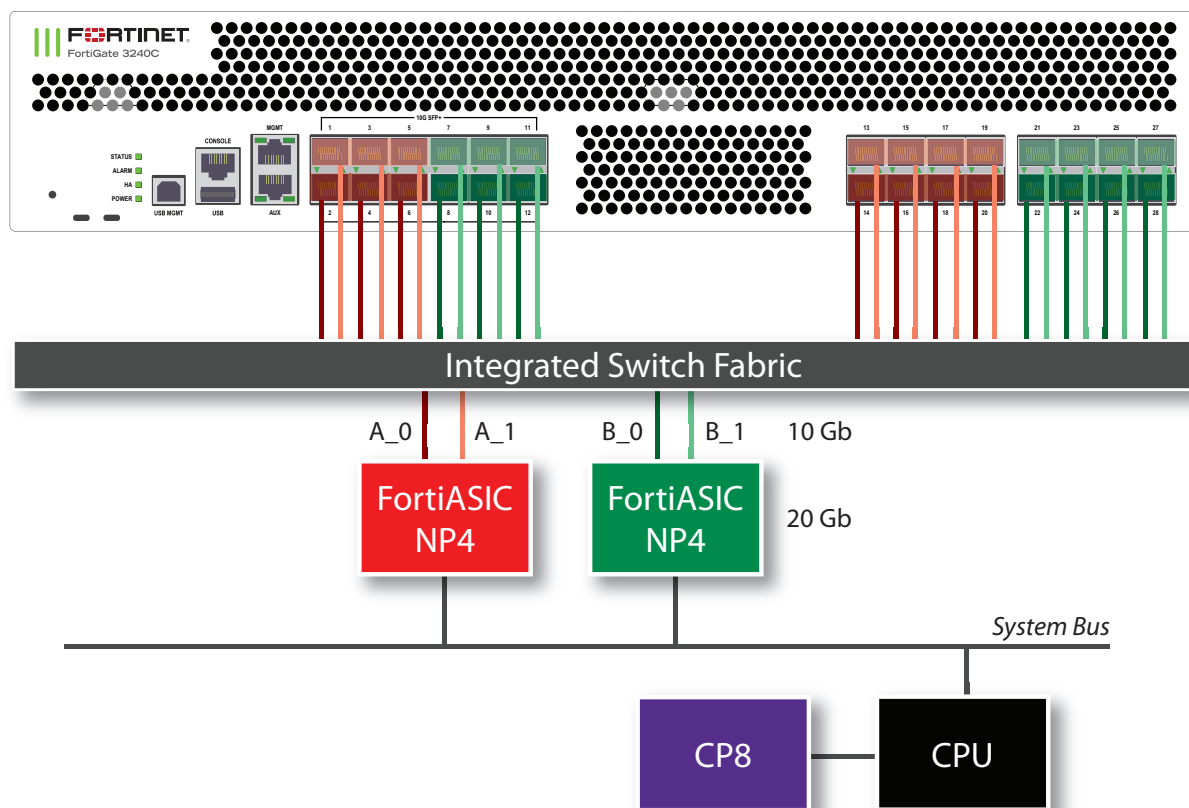
```
config system global
    set sp-load-balance disable
end
```

FortiGate-3240C

The FortiGate-3240C features two NP4 processors:

- The 10Gb interfaces, port1 through port6, and the 1Gb interfaces, port13 through port20, share connections to one NP4 processor.
- The 10Gb interfaces, port7 through port12, and the 1Gb interfaces, port21 through port28, share connections to the other NP4 processor.

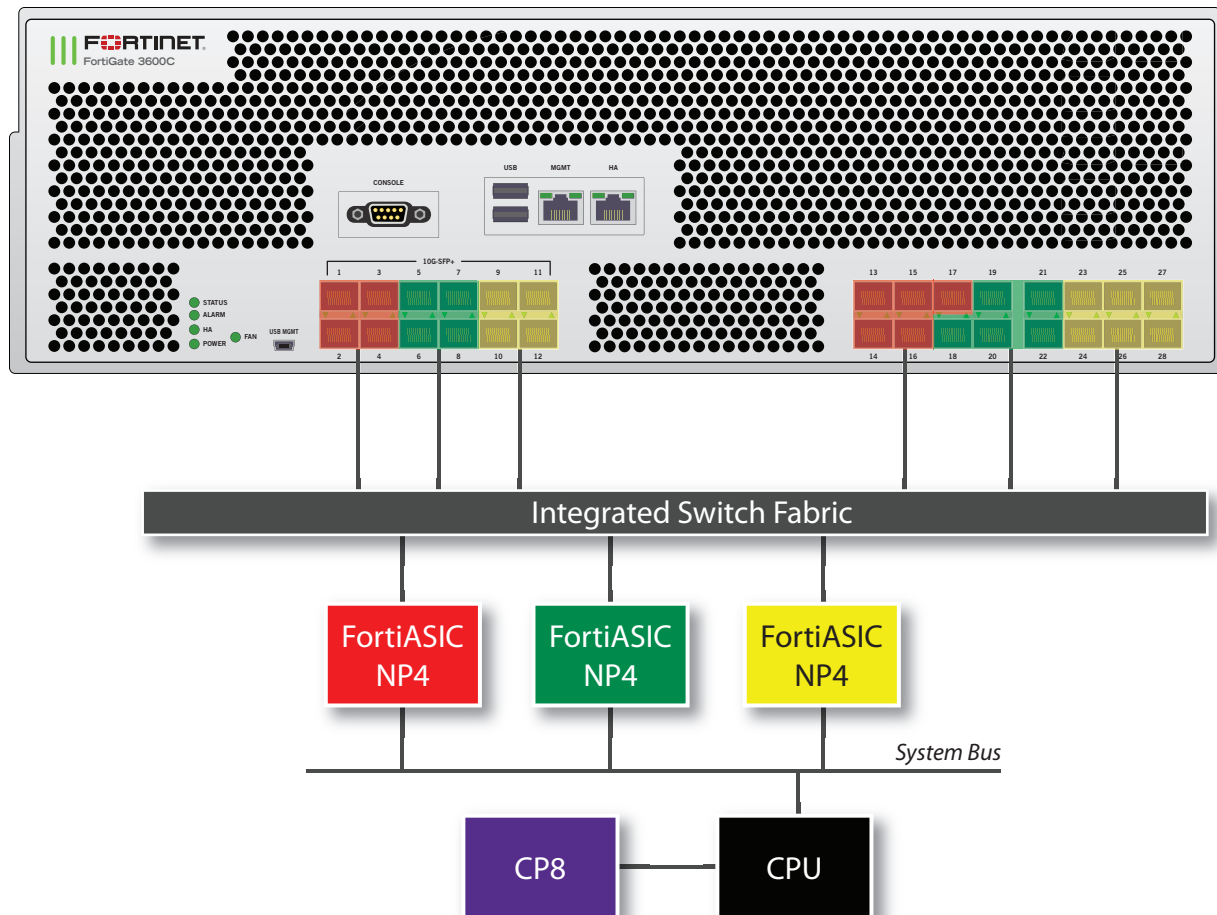
In addition to the ports being divided between the two NP4 processors, they are further divided between the two connections to each processor. Each NP4 can process 20 Gb of network traffic per second and each of two connections to each NP4 can move 10Gb of data to the processor per second, so the ideal configuration would have no more than 10 Gb of network traffic to each connection of each NP4 at any time.



FortiGate-3600C

The FortiGate-3600C features three NP4 processors:

- The 10Gb interfaces, port1-port4, and the 1Gb interfaces, port13-port17, share connections to one NP4 processor.
- The 10Gb interfaces, port5-port8, and the 1Gb interfaces, port18-port22 share connections to the second NP4 processor.
- The 10Gb interfaces, port9-port12, and the 1Gb interfaces, port23-port28 share connections to the third NP4 processor.



XAUI interfaces

Each NP4 processor connects to the integrated switch fabric through two XAUI interfaces: XAUI0 and XAUI1. On each NP4 processor all of the odd numbered interfaces use XAUI0 and all of the even numbered interfaces use XAUI1:

NPU1

XAUI0 = port1, port3, port13, port15, port17

XAUI1 = port2, port4, port14, port16

NPU2

XAUI0 = port5, port7, port18, port20, port22

XAUI1 = port6, port8, port19, port21

NPU3

XAUI0 = port9, port11, port23, port25, port27

XAUI1 = port10, port12, port24, port26, port28

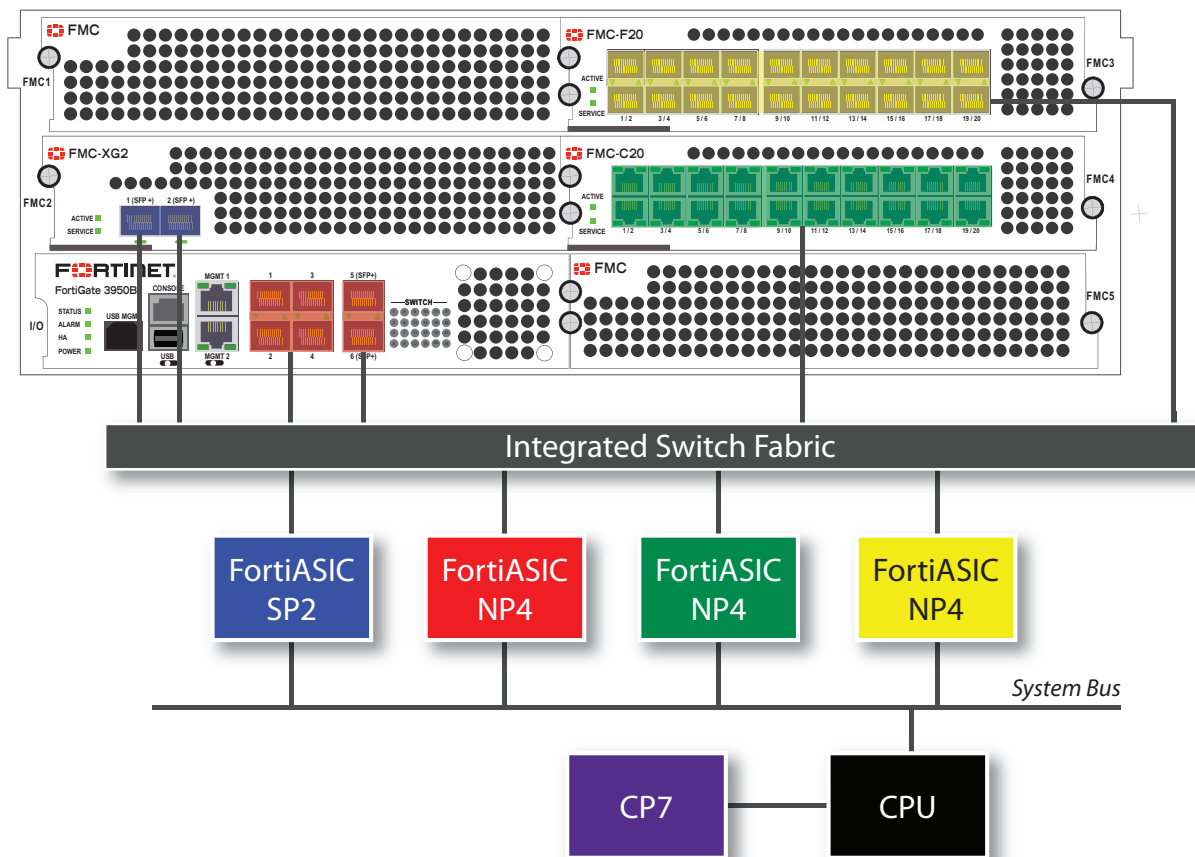
Usually you do not have to be concerned about the XAUI interface mapping. However, if an NP4 interface is processing a very high amount of traffic you should distribute that traffic among both of the XAUI interfaces connected to it. So if you have a very high volume of traffic flowing between two networks you should connect both networks to the same NP4 processor but to different XAUI links. So between even and an odd numbered FortiGate-3600C ports. For example, you could connect one network to port5 and the other network to port6. In this configuration, the second NP4 processor would handle traffic acceleration and both XAUI interfaces would be processing traffic.

FortiGate-3950B and FortiGate-3951B

The FortiGate-3950B features one NP4 processor. The 1Gb SPF interfaces, port1, port2, port3, port4, and the 10Gb SPF+ interfaces, port5, port6, share connections to one NP4 processor. The FortiGate-3951B is similar to the FortiGate-3950B, except it trades one FMC slot for four FSM slots. The network interfaces available on each model are identical.

You can add additional FMC interface modules. The diagram below shows a FortiGate-3950B with three modules installed: an FMC-XG2, an FMC-F20, and an FMC-C20.

- The FMC-XG2 has one SP2 processor. The 10Gb SPF+ interfaces, port1 and port2, share connections to the processor.
- The FMC-F20 has one NP4 processor and the twenty 1Gb SPF interfaces, port1 through port20, share connections to the NP4 processor.
- The FMC-C20 has one NP4 processor and the twenty 10/100/1000 interfaces, port1 through port20, share connections to the NP4 processor.



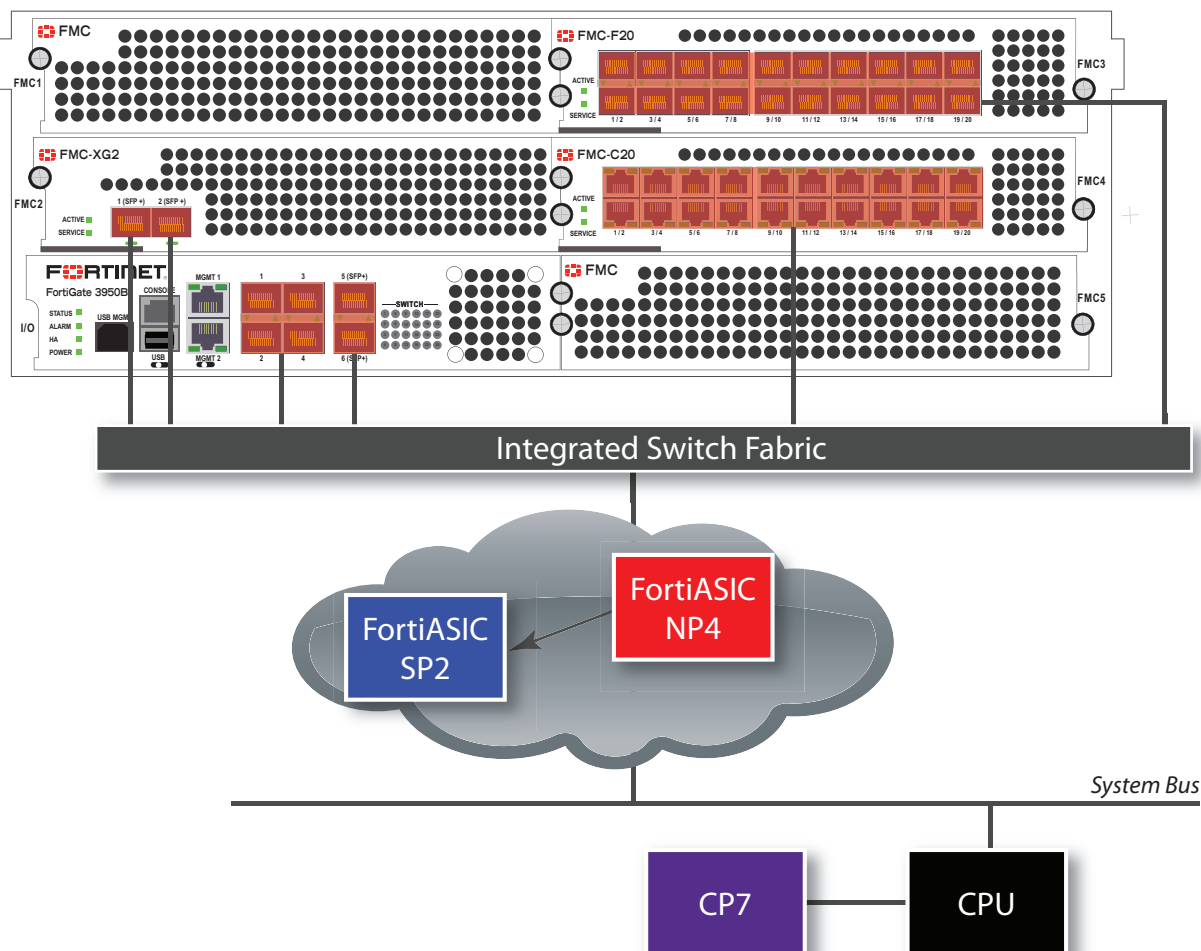
FortiGate-3950B and FortiGate-3951B — load balance mode

Adding one or more FMC-XG2 modules to your FortiGate-3950B allows you to enable load balance mode. This feature allows you increased flexibility in how you use the interfaces on the FortiGate unit. The FortiGate-3951B is similar to the FortiGate-3950B, except it trades one FMC slot for four FSM slots. The network interfaces available on each model are identical.

When enabled, traffic between any two interfaces (excluding management and console) is accelerated whether they are the six interfaces on the FortiGate-3950B itself, or on any installed FMC modules. Traffic is not limited to entering and leaving the FortiGate unit in specific interface groupings to benefit from NP4 and SP2 acceleration. You can use any pair of interfaces.

Security acceleration in this mode is limited, however. Only IPS scanning is accelerated in load balance mode.

The FortiGate-3950B in load balance mode



To enable this feature, issue this CLI command.

```
config system global
    set sp-load-balance enable
```

```
end
```

The FortiGate unit will then restart.

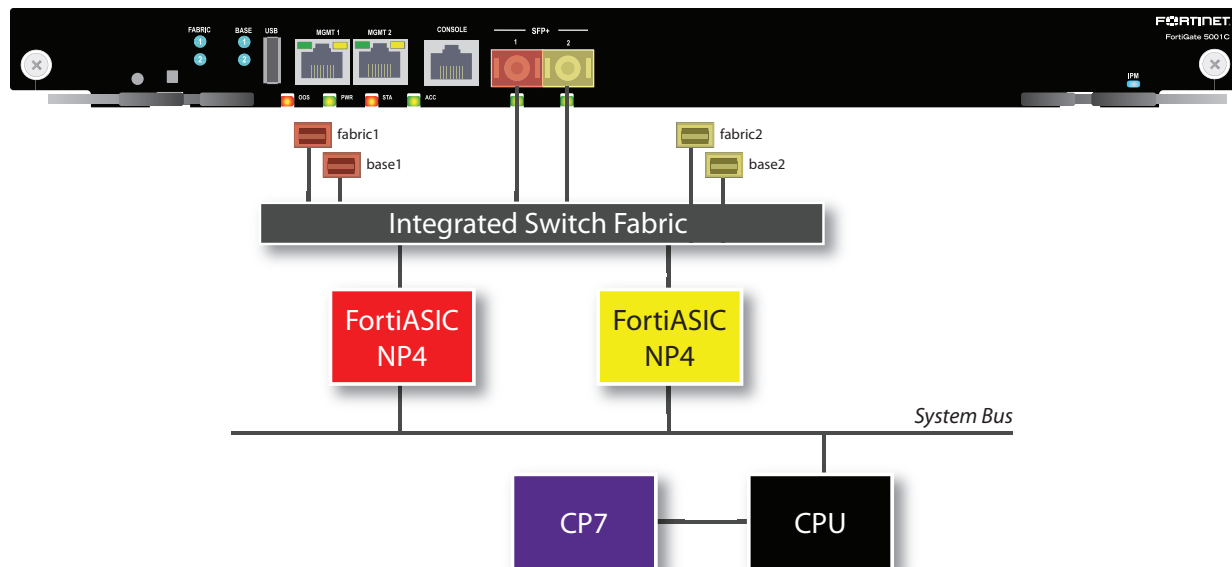
To return to the default mode, issue this CLI command.

```
config system global
    set sp-load-balance disable
end
```

FortiGate-5001C

The FortiGate-5001C board includes two NP4 processors connected to an integrated switch fabric:

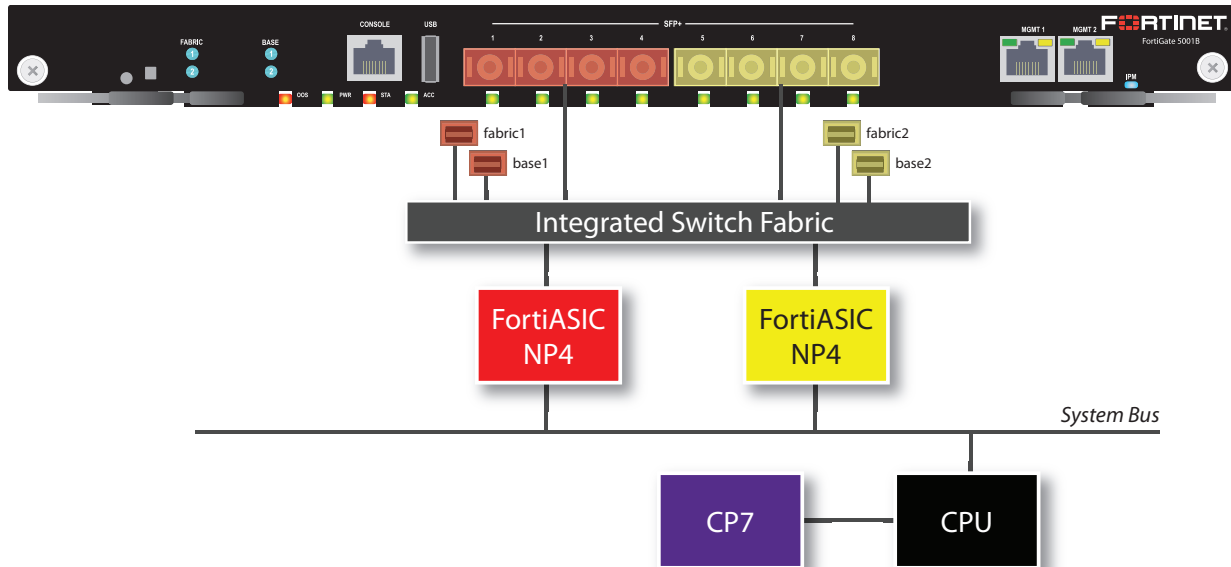
- The port1, fabric1, and base1 interfaces are connected to one NP4 processor.
- The port2, fabric2, and base2 interfaces are connected to the other NP4 processor.



FortiGate-5001B

The FortiGate-5001B board includes two NP4 connected to an integrated switch fabric.

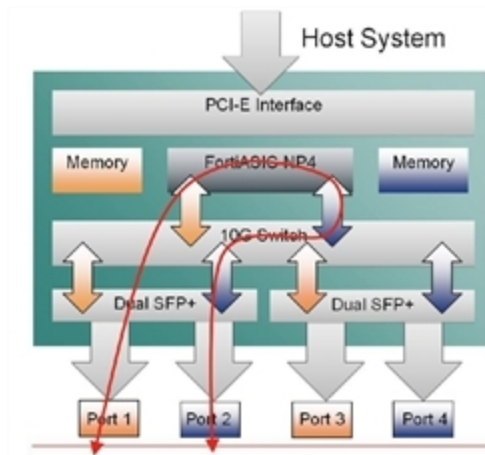
- The port1, port2, port3, port4, fabric1 and base1 interfaces are connected to one NP4 processor.
- The port5, port6, port7, port8, fabric2 and base2 interfaces are connected to the other NP4 processor.



Setting switch-mode mapping on the ADM-XD4

The ADM-XD4 SP has four 10Gb/s ports, but the NP4 processor it contains has only two 10Gb/s ports. The external ports you use are important to optimize the SP for your application.

ADM-XD4 mapping mode



Ports 1 and 3 share one NP4 processor and ports 2 and 4 share the other. Performance ports sharing the same NP4 processor is far better than when forcing network data to move between NP4 processors by using one port from each, for example ports 1 and 2 or ports 3 and 4.

Hardware acceleration diagnose commands

This section describes some diagnose commands you can use to display useful information about sessions processed by NP6 and other NPx processors.

diagnose sys session/session6 list (IPv4, IP6, NAT46, NAT64, and IPsec session information)

Display information about the IPv4, IPv6, NAT46, NAT64 and IPsec sessions being processed by NPx processors. The command output includes the following:

- offload=1/1 for NP1(FA1) sessions.
- offload=2/2 for NP1(FA2) sessions.
- offload=3/3 for NP2 sessions.
- offload=4/4 for NP4 sessions.
- offload=5/5 for XLR sessions.
- offload=6/6 for Nplite/NP4lite sessions.
- offload=7/7 for XLP sessions.
- offload=8/8 for NP6 sessions.
- flag 0x81 means regular traffic.
- flag 0x82 means IPsec traffic.

Example IPv4 NP6 session

```
diag sys session list
session info: proto=6 proto_state=01 duration=2 expire=3597 timeout=3600 flag-
s=00000000 sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/
state=may_dirty npu
statistic(bytes/packets/allow_err): org=112/2/1 reply=60/1/1 tuples=2
speed(Bps/kbps): 0/0
origin->sink: org pre->post, reply pre->post dev=13->14/14->13 gwy-
y=10.1.100.11/172.16.200.55
hook=pre dir=org act=noop 172.16.200.55:57975->10.1.100.11:80(0.0.0.0:0)
hook=post dir=reply act=noop 10.1.100.11:80->172.16.200.55:57975(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0
serial=04051918 tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
npu_state=0x000c00
```



```
npu info: flag=0x81/0x81, offload=8/8, ips_offload=0/0, epid=137/136, ipid-  
d=136/137, vlan=0/0
```

Example IPv4 IPsec NP6 session

```
diag sys session list  
session info: proto=6 proto_state=01 duration=34 expire=3565 timeout=3600 flag-  
s=00000000 sockflag=00000000 sockport=0 av_idx=0 use=3  
origin-shaper=  
reply-shaper=  
per_ip_shaper=  
ha_id=0 policy_dir=0 tunnel=/pl-vdom2  
state=re may_dirty npu  
statistic(bytes/packets/allow_err): org=112/2/1 reply=112/2/1 tuples=2  
origin->sink: org pre->post, reply pre->post dev=57->7/7->57 gwy-  
y=10.1.100.11/11.11.11.1  
hook=pre dir=org act=noop 172.16.200.55:35254->10.1.100.11:80(0.0.0.0:0)  
hook=post dir=reply act=noop 10.1.100.11:80->172.16.200.55:35254(0.0.0.0:0)  
pos/(before,after) 0/(0,0), 0/(0,0)  
misc=0 policy_id=1 id_policy_id=0 auth_info=0 chk_client_info=0 vd=4  
serial=00002d29 tos=ff/ff ips_view=0 app_list=0 app=0  
dd_type=0 dd_mode=0  
per_ip_bandwidth meter: addr=172.16.200.55, bps=260  
npu_state=00000000  
npu info: flag=0x81/0x82, offload=8/8, ips_offload=0/0, epid=1/3, ipid=3/1, vlan-  
n=32779/0
```

Example IPv6 NP6 session

```
diag sys session6 list  
session6 info: proto=6 proto_state=01 duration=2 expire=3597 timeout=3600 flag-  
s=00000000 sockport=0 sockflag=0 use=3  
origin-shaper=  
reply-shaper=  
per_ip_shaper=  
ha_id=0  
policy_dir=0 tunnel=/  
state=may_dirty npu  
statistic(bytes/packets/allow_err): org=152/2/0 reply=152/2/0 tuples=2  
speed(Bps/kbps): 0/0  
origin->sink: org pre->post, reply pre->post dev=13->14/14->13  
hook=pre dir=org act=noop 2000:172:16:200::55:59145 ->2000:10:1:100::11:80(:::0)  
hook=post dir=reply act=noop 2000:10:1:100::11:80 ->2000:172:16:200::55:59145  
(:::0)  
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0 serial=0000027a  
npu_state=0x000c00  
npu info: flag=0x81/0x81, offload=8/8, ips_offload=0/0, epid=137/136, ipid-  
d=136/137, vlan=0/0
```

Example NAT46 NP6 session

```
diagnose sys session list
session info: proto=6 proto_state=01 duration=19 expire=3580 timeout=3600 flag-
s=000000000 sockflag=000000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/
state=npu nlb
statistic(bytes/packets/allow_err): org=112/2/1 reply=112/2/1 tuples=2
speed(Bps/kbps): 0/0
origin->sink: org nataf->post, reply pre->org dev=52->14/14->52 gwy-
y=0.0.0.0/10.1.100.1
hook=5 dir=org act=noop 10.1.100.1:21937->10.1.100.11:80(0.0.0.0:0)
hook=6 dir=reply act=noop 10.1.100.11:80->10.1.100.1:21937(0.0.0.0:0)
hook=pre dir=org act=noop 2000:172:16:200::55:33945 ->64:ff9b::a01:640b:80(:::0)
hook=post dir=reply act=noop 64:ff9b::a01:640b:80 ->2000:172:16:200::55:33945
(:::0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0
serial=04051aae tos=ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
npu_state=000000000
npu info: flag=0x81/0x00, offload=0/8, ips_offload=0/0, epid=0/136, ipid=0/137,
vlan=0/0
```

Example NAT64 NP6 session

```
diagnose sys session6 list
session6 info: proto=6 proto_state=01 duration=36 expire=3563 timeout=3600 flag-
s=000000000 sockport=0 sockflag=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0
policy_dir=0 tunnel=/
state=may_dirty npu nlb
statistic(bytes/packets/allow_err): org=72/1/0 reply=152/2/0 tuples=2
speed(Bps/kbps): 0/0
origin->sink: org pre->org, reply nataf->post dev=13->14/14->13
hook=pre dir=org act=noop 2000:172:16:200::55:33945 ->64:ff9b::a01:640b:80(:::0)
hook=post dir=reply act=noop 64:ff9b::a01:640b:80 ->2000:172:16:200::55:33945
(:::0)
hook=5 dir=org act=noop 10.1.100.1:21937->10.1.100.11:80(0.0.0.0:0)
hook=6 dir=reply act=noop 10.1.100.11:80->10.1.100.1:21937(0.0.0.0:0)
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0 serial=0000027b
npu_state=000000000
npu info: flag=0x00/0x81, offload=8/0, ips_offload=0/0, epid=137/0, ipid=136/0,
vlan=0/0
```

diagnose npu np6 session-stats <np6-id> (number of NP6 IPv4 and IPv6 sessions)

You can use the `diagnose npu np6 portlist` command to list the NP6-ids and the interfaces that each NP6 is connected to. The <np6-id> of np6_0 is 0, the <np6-id> of np6_1 is 1 and so on. The `diagnose npu np6 session-stats <np6-id>` command output includes the following headings:

- `ins44` installed IPv4 sessions
- `ins46` installed NAT46 sessions
- `del4` deleted IPv4 and NAT46 sessions
- `ins64` installed NAT64 sessions
- `ins66` installed IPv6 sessions
- `del6` deleted IPv6 and NAT64 sessions
- `e` is the error counter for each session type

`diagnose npu np6 session-stats 0`

qid	ins44	ins46	del4	ins64	ins66	del6
	ins44_e	ins46_e	del4_e	ins64_e	ins66_e	del6_e

0	94	0	44	0	40	30
	0	0	0	0	0	0
1	84	0	32	0	30	28
	0	0	0	0	0	0
2	90	0	42	0	40	30
	0	0	0	0	0	0
3	86	0	32	0	24	27
	0	0	0	0	0	0
4	72	0	34	0	34	28
	0	0	0	0	0	0
5	86	0	30	0	28	32
	0	0	0	0	0	0
6	82	0	38	0	32	34
	0	0	0	0	0	0
7	86	0	30	0	30	30
	0	0	0	0	0	0
8	78	0	26	0	36	26
	0	0	0	0	0	0
9	86	0	34	0	32	32
	0	0	0	0	0	0

Total	844	0	342	0	326	297
	0	0	0	0	0	0

diagnose npu np6 ipsec-stats (NP6 IPsec statistics)

The command output includes IPv4, IPv6, and NAT46 IPsec information:

- spi_ses4 is the IPv4 counter
- spi_ses6 is the IPv6 counter
- 4to6_ses is the NAT46 counter

```
diagnose npu np6 ipsec-stats
vif_start_oid      03ed      vif_end_oid      03fc
IPsec Virtual interface stats:
vif_get            000000000000      vif_get_expired  000000000000
vif_get_fail      000000000000      vif_get_invld    000000000000
vif_set           000000000000      vif_set_fail     000000000000
vif_clear         000000000000      vif_clear_fail   000000000000
np6_0:
sa_install        000000000000      sa_ins_fail      000000000000
sa_remove         000000000000      sa_del_fail      000000000000
4to6_ses_ins      000000000000      4to6_ses_ins_fail 000000000000
4to6_ses_del      000000000000      4to6_ses_del_fail 000000000000
spi_ses6_ins      000000000000      spi_ses6_ins_fail 000000000000
spi_ses6_del      000000000000      spi_ses6_del_fail 000000000000
spi_ses4_ins      000000000000      spi_ses4_ins_fail 000000000000
spi_ses4_del      000000000000      spi_ses4_del_fail 000000000000
sa_map_alloc_fail 000000000000      vif_alloc_fail   000000000000
sa_ins_null_adapter 000000000000      sa_del_null_adapter 000000000000
del_sa_mismatch   000000000000      ib_chk_null_adpt 000000000000
ib_chk_null_sa    000000000000      ob_chk_null_adpt 000000000000
ob_chk_null_sa    000000000000      rx_vif_miss      000000000000
rx_sa_miss        000000000000      rx_mark_miss     000000000000
waiting_ib_sa     000000000000      sa_mismatch      000000000000
msg_miss          000000000000
np6_1:
sa_install        000000000000      sa_ins_fail      000000000000
sa_remove         000000000000      sa_del_fail      000000000000
4to6_ses_ins      000000000000      4to6_ses_ins_fail 000000000000
4to6_ses_del      000000000000      4to6_ses_del_fail 000000000000
spi_ses6_ins      000000000000      spi_ses6_ins_fail 000000000000
spi_ses6_del      000000000000      spi_ses6_del_fail 000000000000
spi_ses4_ins      000000000000      spi_ses4_ins_fail 000000000000
spi_ses4_del      000000000000      spi_ses4_del_fail 000000000000
sa_map_alloc_fail 000000000000      vif_alloc_fail   000000000000
sa_ins_null_adapter 000000000000      sa_del_null_adapter 000000000000
del_sa_mismatch   000000000000      ib_chk_null_adpt 000000000000
ib_chk_null_sa    000000000000      ob_chk_null_adpt 000000000000
ob_chk_null_sa    000000000000      rx_vif_miss      000000000000
rx_sa_miss        000000000000      rx_mark_miss     000000000000
waiting_ib_sa     000000000000      sa_mismatch      000000000000
msg_miss          000000000000
```

diagnose sys mcast-session/session6 list (IPv4 and IPv6 multicast sessions)

This command lists all IPv4 or IPv6 multicast sessions. If a multicast session can be offloaded, the output includes the `offloadable` tag. If the multicast path can be offloaded one of the paths in the command output is tagged as `offloaded`.

The only way to determine the number of offloaded multicast sessions is to use the `diagnose sys mcast-session/session6 list` command and count the number of sessions with the `offload` tag.

```
diagnose sys mcast-session list
session info: id=3 vf=0 proto=17 172.16.200.55.51108->239.1.1.1.7878
used=2 path=11 duration=1 expire=178 indev=6 pkts=2 state:2cpu offloadable
npu-info in-pid=0 vifid=0 in-vtag=0 npuid=0 queue=0 tae=0

path: 2cpu policy=1, outdev=2
      out-vtag=0
path: 2cpu policy=1, outdev=3
      out-vtag=0
path: offloaded policy=1, outdev=7
      out-vtag=0
path: policy=1, outdev=8
      out-vtag=0
path: policy=1, outdev=9
      out-vtag=0
path: policy=1, outdev=10
      out-vtag=0
path: policy=1, outdev=11
      out-vtag=0
path: policy=1, outdev=12
      out-vtag=0
path: policy=1, outdev=13
      out-vtag=0
path: 2cpu policy=1, outdev=64
      out-vtag=0
path: 2cpu policy=1, outdev=68
      out-vtag=0
```

diagnose npu np6 sse-stats <np6-id> (number of NP6 sessions and dropped sessions)

This command displays the total number of inserted, deleted and purged sessions processed by a selected NP6 processor. The number of dropped sessions of each type can be determined by subtracting the number of successful sessions from the total number of sessions. For example, the total number of dropped insert sessions is `insert-total - insert-success`.

```
diagnose npu np6 sse-stats 0
```

Counters	SSE0	SSE1	Total
active	0	0	0
insert-total	25	0	0
insert-success	25	0	0
delete-total	25	0	0
delete-success	25	0	0
purge-total	0	0	0
purge-success	0	0	0
search-total	40956	38049	79005
search-hit	37714	29867	67581
pht-size	8421376	8421376	
oft-size	8355840	8355840	
oftfree	8355839	8355839	
PBA	3001		

diagnose npu np6 dce <np6-id> (number of dropped NP6 packets)

This command displays the number of dropped packets for the selected NP6 processor.

- IHP1_PKTCHK number of dropped IP packets
- IPSEC0_ENGINB0 number of dropped IPSec
- TPE_SHAPER number of dropped traffic sharper packets

```
diag npu np6 dce 1
```

```
IHP1_PKTCHK :00000000000001833 [5b] IPSEC0_ENGINB0 :00000000000000003 [80]
TPE_SHAPER :00000000000000552 [94]
```

diagnose hardware deviceinfo nic <interfac-name> (number of packets dropped by an interface)

This command displays a wide variety of statistics for FortiGate interfaces. The fields `Host Rx dropped` and `Host Tx dropped` display the number of received and transmitted packets that have been dropped.

```
diagnose hardware deviceinfo nic port2
```

```
...
```

```
===== Counters =====
```

```
Rx Pkts           :20482043
Rx Bytes          :31047522516
Tx Pkts           :19000495
Tx Bytes          :1393316953
Host Rx Pkts      :27324
Host Rx Bytes     :1602755
Host Rx dropped   :0
Host Tx Pkts      :8741
Host Tx Bytes     :5731300
Host Tx dropped   :0
sw_rx_pkts        :20482043
sw_rx_bytes       :31047522516
sw_tx_pkts        :19000495
sw_tx_bytes       :1393316953
sw_np_rx_pkts     :19000495
sw_np_rx_bytes    :1469318933
sw_np_tx_pkts     :20482042
sw_np_tx_bytes    :31129450620
```

diagnose npu np6 synproxy-stats (NP6 SYN-proxied sessions and unacknowledged SYNs)

This command displays information about NP6 syn-proxy sessions including the total number proxied sessions. As well the Number of attacks, no ACK from client shows the total number of unacknowledged SYNs.

```
diagnose npu np6 synproxy-stats
DoS SYN-Proxy:
Number of proxied TCP connections : 39277346
Number of working proxied TCP connections : 182860
Number of retired TCP connections : 39094486
Number of attacks, no ACK from client : 208
```




High Performance Network Security



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